

Intrinsic-Dimension-Guided Bottleneck Selection for Autoencoders and Variational Autoencoders with Active-Unit Diagnostics

Chika Obata ¹ , Mizuki Dai ¹ , and Kenya Jin'no ^{1,*} 

¹ Department of Intelligent Systems, Faculty of Information Engineering, Tokyo City University, 1-28-1 Tamazutsumi, Setagaya-ku, Tokyo 158-8557, Japan; g2332016@tcu.ac.jp (C.O.); g2591402@tcu.ac.jp (M.D.)

* Correspondence: kjinno@tcu.ac.jp; Tel.: +81-3-5707-0104 (K.J.)

Abstract

The bottleneck dimension m is a fundamental design choice in autoencoders (AEs) and variational autoencoders (VAEs), yet in practice it is usually selected by convention or trial and error. We propose a **conditional, diagnostic protocol** that uses an estimate $\hat{d}_{ID}$ of the intrinsic dimension of the input data as a data-dependent **search-window reference** for selecting m . For AEs, the protocol recommends searching within a finite window centered on $\hat{d}_{ID}$, typically $m \in [\hat{d}_{ID}, 2\hat{d}_{ID}]$; for VAEs, it recommends selecting a sufficiently generous candidate dimension $m \geq 2\hat{d}_{ID}$ and verifying after training whether the surplus coordinates became inactive, using active units (AU) and reconstruction error. The geometric motivation is a topological lower bound for zero-error continuous reconstruction of an idealized manifold; the VAE-side interpretation is supported by a linear-Gaussian rate-distortion surrogate with activation condition $\lambda_j > \beta\tau^2$. On synthetic manifolds with known ground truth (three to five seeds) and on fully connected VAEs trained on MNIST and Fashion-MNIST (three seeds each), the protocol produced reproducible diagnostics, using a pre-specified Step-3 screening rule for VAEs together with complementary reconstruction-error diagnostics (MNIST passes; on Fashion-MNIST the rule diagnostically detects an over-pruning boundary case), and for the specific MNIST grid and evaluation protocol used here the ID-guided window reduces the training runs of a full grid search by roughly 60–80% while containing the best observed linear-probe accuracy. The method is conditional rather than universal: under the datasets, architectures, and estimators tested, intrinsic-dimension estimates were numerically stable only through reference representations with $m \gg \hat{d}_{ID}$ —on natural images, latent-space TwoNN estimates increased when the reference bottleneck exceeded the provisional consistency region—and for the tested Conv-VAE schedule with $\beta_{\max} = 4$, AU-based truncation was not observed within the tested range, whereas increasing $\beta_{\max}$ reduced the activation ratio (3 seeds) without establishing order consistency. These results support using $\hat{d}_{ID}$ as a search-window reference and AU as a **model- and training-dependent diagnostic**—not as general estimators of intrinsic or embedding dimension.

Keywords: variational autoencoder; bottleneck dimension; intrinsic dimension; manifold hypothesis; active units; TwoNN; diagnostic protocol; model selection; representation learning; knowledge extraction

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1. Introduction

1.1. Background: How Should the Bottleneck Dimension Be Chosen?

Autoencoders (AEs) and variational autoencoders (VAEs) [1] are core unsupervised models that learn low-dimensional latent representations from high-dimensional data [2]. Their most fundamental design parameter is the bottleneck dimension m : if m is too small, information is lost; if it is too large, redundant dimensions arise and, in VAEs, complicate the interpretation of posterior collapse [3]. Nevertheless, in practice m is usually set to “a round number such as 32 or 64” or tuned by trial and error, and a selection principle grounded in the data itself has not been established.

On the other hand, according to the manifold hypothesis [4], natural data concentrate near a low-dimensional manifold inside the high-dimensional input space $\mathbb{R}^N$, and the local number of degrees of freedom is characterized by the intrinsic dimension $d_{\text{ID}} \ll N$. Indeed, estimators such as TwoNN [5] report that the intrinsic dimension of natural images is far smaller than the pixel dimension [6]. From this viewpoint, **the appropriate scale of the bottleneck dimension m should be informed not by model-side convention but by a data-side geometric quantity—namely, an estimate of the local intrinsic dimensionality.** The question addressed in this paper is:

Can the bottleneck dimension m of an AE/VAE be chosen in a principled way from an estimate of the intrinsic (manifold) dimension of the input data? And is there a means of verifying, after training, that the choice was appropriate?

1.2. Central Idea and Contributions

The central claim of this paper can be stated in one sentence:

An estimate $\hat{d}_{\text{ID}}$ of the intrinsic dimension of the input data provides a search-window reference for the bottleneck dimension m —“AE: $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$; VAE: a generous candidate $m \geq 2\hat{d}_{\text{ID}}$ ”—and the active units (AU) of a VAE—the number of coordinates actually used under KL regularization—serve, under a suitable β and architecture, as a post-training diagnostic of whether the surplus dimensions were actually truncated.

This is not a claim of a general solution to choosing m , nor a single-number selection algorithm; it is the proposal of a **conditional diagnostic protocol** (applicability conditions are summarized in Table 3). In particular, $\hat{d}_{\text{ID}}$ provides the center or lower side of a candidate range rather than a unique correct answer, the VAE-side rule $m \geq 2\hat{d}_{\text{ID}}$ is neither a sufficient condition nor an optimality principle (in practice the candidate is picked from a finite pre-specified grid), and AU is not an estimator of intrinsic or embedding dimension but a diagnostic that depends on β , the AU threshold, the architecture, and the training conditions. Within the linear-Gaussian surrogate of Section 4, the activation condition $\lambda_j > \beta\tau^2$ (Proposition 2) predicts that even if m is set above the required dimension, the KL regularization of the VAE truncates the surplus dimensions (posterior collapse) and the AU count reports the effective number of used dimensions—in other words, **whereas for an AE the choice of m itself fixes the representation dimension, for a VAE the conservative operating strategy of using a generous candidate dimension followed by AU-based verification may be useful when KL-induced truncation is effective**—a condition that is itself a hypothesis to be checked by the Step-3 decision rule.

The contributions and theoretical motivation of this paper are organized as follows:

1. **Methodological contribution: an ID-guided bottleneck-dimension selection protocol** (Section 3): an estimate-set-verify workflow consisting of (S1) estimating $\hat{d}_{\text{ID}}$ (TwoNN/MLE; for real data through a reference AE with $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$), (S2) setting m (AE: $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$ plus a topological margin; VAE: a generous setting $m \geq 2\hat{d}_{\text{ID}}$),

- and (S3) post-training verification by AU and reconstruction error, together with a **reproducible pass/fail decision rule** (Equations (2)–(3)) and reporting requirements.
2. **Diagnostic contribution:** a failure-mode taxonomy (Table 2) that combines AU and MSE to distinguish surplus-dimension truncation, over-regularization, Step-1 underestimation, and topological obstructions, demonstrated on ground truth (Torus over-pruning caught by the MSE cross-check) and on real data (Fashion-MNIST lower-V2-bound violation).
 3. **Empirical contribution: multi-seed validation and quantified applicability boundaries** (Section 5): the full protocol is validated on synthetic manifolds with known ground truth (3–5 seeds) and on fully connected VAEs on MNIST (3 seeds; $\hat{d}_{ID} \approx 10$ –12 consistent with the AU slowdown region), with Step-1 stability supported by sub-sampling confidence intervals, cross-checks with estimators of different principles (DANCo, ESS), and a 3-seed reference-model comparison (Section 5.5); a linear probe shows that on the specific MNIST grid tested here the ID-guided range reduces search cost while containing the best observed downstream accuracy (Section 5.4). The applicability limits are quantified in the same experiments: (a) under the tested datasets, architectures, and estimators, $\hat{d}_{ID}$ was numerically stable only through reference representations with $m \gg \hat{d}_{ID}$, while isometry improvements were secondary (Section 5.5); (b) on natural images the TwoNN estimate agrees with reported values [6] up to $m \approx 2\hat{d}_{ID}$ but inflates beyond; (c) for Conv-VAEs at $\beta_{\max} = 4$ no substantial truncation is observed within $m \leq 512$ —consistent, under the surrogate interpretation, with an effective truncation threshold beyond the tested range—so AU verification requires a strengthened β (the β -strengthened recovery achieves V1 only, not V2; 3 seeds), and Step 3 on natural images remains unresolved—the contributions are claimed inclusive of these limitations.
 4. **Theoretical motivation** (Section 4): an idealized topological lower bound for zero-error continuous reconstruction (Proposition 1) and the surrogate activation condition $\lambda_j > \beta\tau^2$ under a linear-Gaussian rate–distortion approximation (Proposition 2). Both are motivations rather than guarantees for trained nonlinear networks: the lower bound assumes zero reconstruction error and continuity, and the activation condition is exact only for the linear-Gaussian surrogate (its scope is stated in the claim bodies).

Relation to prior work: studies that “measure” the intrinsic dimension of deep representations [6–8] stop at analyzing trained networks, whereas this work uses intrinsic-dimension estimation for the **design of the bottleneck dimension** and formalizes a closed-loop verification of that design through AU. Whereas the linear analysis of posterior collapse [3] remains qualitative, we organize it into the closed-form threshold $\lambda_j > \beta\tau^2$ (the novelty lies in the integrated organization; Section 4) and operate it as the rationale of the selection protocol.

1.3. Two Faces of the “Manifold Dimension”: d_{ID} and d_{emb}

As the “manifold dimension of the input data” we distinguish two quantities: the **local intrinsic dimension** d_{ID} (the local degrees of freedom of the manifold; the target of TwoNN/MLE) and the **topological minimum embedding dimension** d_{emb} (the minimum dimension for embedding into Euclidean space without self-intersection). For the Swiss Roll $d_{emb} = d_{ID} = 2$, but for the Torus $d_{emb} = 3 > d_{ID} = 2$. What the bottleneck requires is d_{emb} (Proposition 1); since only d_{ID} can be estimated for real data, this gap (topological obstruction) is one reason for the margin “set m above $\hat{d}_{ID}$ ” (Section 3.2.2). One caution applies throughout: for an ideal d -dimensional smooth manifold the minimum Euclidean embedding dimension satisfies $d_{emb} \geq d$, but in real datasets $\hat{d}_{ID}$ is an estimator of **local intrinsic dimensionality**, not a direct estimator of d_{emb} . The relation between intrinsic and

embedding dimensions should therefore be read as a **geometric motivation for adding a margin**, not as an empirically verifiable inequality involving $\hat{d}_{\text{ID}}$.

1.4. Limitations of This Paper (Summary)

Our validation covers synthetic manifolds with known ground truth ($d_{\text{true}} \leq 3$), MNIST and Fashion-MNIST (fully connected AE/VAE, 3 seeds), and—as quantification of the applicability boundary—Conv-VAEs and natural images (CIFAR-10, SVHN). The state of validation is asymmetric: **fully connected architectures receive complete multi-seed validation**, whereas **convolutional architectures and natural images are at the stage of probing the applicability boundary** (a 3-seed β -strengthening experiment and single-seed exploratory observations). Two further caveats recur throughout: a numerically stable $\hat{d}_{\text{ID}}$ is not necessarily unbiased (stability vs. validity; Section 6.2), so $\hat{d}_{\text{ID}}$ is operated as a **reference value**, not as the true value; and automatic change-point (knee) detection on the AU curve is strongly grid- and seed-dependent, so it is not a named protocol output (exploratory analysis in Appendix B). Full application to large-scale natural images remains future work; a detailed discussion of limitations is given in Section 6.4.

The remainder of the paper covers related work (Section 2), the protocol (Section 3), its theoretical motivation (Section 4), experiments (Section 5), discussion (Section 6), and conclusions (Section 7). Proofs, sensitivity analyses, and exploratory experiments are collected in the appendices.

2. Related Work

Representation learning and autoencoders: Bengio et al. [2] organized the framework in which AEs learn the low-dimensional structure of high-dimensional data based on the manifold hypothesis [4]. The β -VAE [9] controls the degree of compression of the representation through the KL weight β , which directly affects AU. DAE [10] and CAE [11] are regularized variants that promote learning of manifold structure; in this paper they serve as candidate reference representations for intrinsic-dimension estimation (Section 5.5).

Intrinsic-dimension analysis of deep representations: Pope et al. [6] showed that the intrinsic dimension of natural images is far lower than the pixel dimension (CIFAR-10: $d_{\text{ID}} \approx 26$), and Ansuini et al. [7] quantified how the ID changes across layers of deep networks. Valeriani et al. [8] analyzed the hidden-representation geometry of large transformers, and Birdal et al. [12] related ID to generalization. These are studies that “measure” trained models; our work is complementary in that it uses the measurements for the **design and verification of the bottleneck dimension**. As ID estimators we adopt TwoNN [5] and MLE [13]; DANCo [14] can provide a useful complementary estimator but is computationally more expensive, so we do not use it as a primary estimator and instead apply it as an auxiliary robustness cross-check on selected MNIST latents (Section 5.3.2).

Limitations of intrinsic-dimension estimators: nearest-neighbor ID estimators are known to suffer from finite-sample bias, sensitivity to non-uniform sampling, noise, and curvature, and from distance concentration when the ambient dimension is large relative to the sample size (see the survey by Camastra and Staiano [15]). Real data may moreover be better described as a union of submanifolds with heterogeneous local intrinsic dimension, which conflicts with the single-manifold assumption underlying global estimators [16], and estimates computed on deep representations depend on the representation itself [6,7]. Our experiments encounter all of these effects (sample-size-dependent bias in Section 5.3.2; representation dependence in Sections 5.5 and 5.6), which is why the protocol treats $\hat{d}_{\text{ID}}$ as a conditional reference value with mandatory convergence and cross-estimator checks rather than as a trusted point estimate.

Active units and posterior collapse: AU [3,9] is the number of latent dimensions whose posterior-mean variance exceeds a threshold δ , and measures the number of coordinates actually used under KL regularization. Lucas et al. [3] analyzed the mechanism of posterior collapse in linear VAEs. This work incorporates that mechanism into the selection protocol as a “safety device that can truncate an over-generous m ” (whether truncation actually occurred is diagnosed in Step 3), with the water-filling activation condition as its theoretical rationale (Proposition 2; the PPCA of Tipping & Bishop [17] is the special case $\beta = 1$).

Automatic determination of the number of latent dimensions: within linear and probabilistic models, Bayesian PCA with an ARD (automatic relevance determination) prior [18] determines the number of principal components, and Bayesian nonparametric latent variable models such as those based on the Indian Buffet Process [19] determine the number of latent features, during learning. The KL truncation of the β -VAE performs a similar “automatic pruning of surplus dimensions” in the nonlinear setting, but the pruning result depends on β and the architecture (Section 5.6). Rather than relying on model-side automatic determination, our protocol derives a search window from the data-geometry side ($\hat{d}_{\text{ID}}$) and verifies it against the pruning result (AU); the two approaches are complementary.

Geometric autoencoders: Nazari et al. [20] and Kato et al. [21] proposed regularizers that preserve the geometry (isometry) of the latent space. We compare the effect of isometry on the stability of ID estimation against the “bottleneck margin $m/\hat{d}_{\text{ID}}$ ” and show that the latter is dominant (Section 5.5).

3. ID-Guided Protocol for Selecting the Bottleneck Dimension

3.1. Problem Setting and Main Symbols

We define an AE/VAE with input dimension N , bottleneck dimension m , encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$, and decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. In a VAE the encoder outputs a posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$, and the KL regularization (strength β) acts as an information bottleneck.

We define the **active units (AU)** as

$$\text{AU}(m) = \sum_{k=1}^m \mathbf{1}[\text{Var}_x[\mu_k(x)] > \delta], \quad \delta = 10^{-2}, \quad (1)$$

where $\mu_k(x)$ is the k -th component of the posterior mean; the threshold δ follows Lucas et al. [3], and we have confirmed that AU values are essentially unchanged for $\delta \in [10^{-3}, 10^{-1}]$ (Appendix D). AU is computed on test data (the train/test distinction is stated in each experiment section). Three definitional remarks on AU apply throughout. First, AU is based on the data variance of the **posterior mean**; it is not identical to a per-coordinate decomposition of the posterior variance or of the KL contribution. Second, the threshold δ is compared against latent-coordinate variances and can in principle depend on the latent scale (in our settings the standard-normal prior anchors the scale, and the δ -insensitivity above holds). Third, **AU is coordinate-dependent**: it is not invariant under arbitrary rotations or reparameterizations of the latent space, so its interpretation depends on the chosen prior, parameterization, training objective, and architecture.

Table 1. Main symbols. Each quantity is distinguished so that the protocol’s claims are stated without conflating targets and estimators.

Symbol	Meaning	Role in the protocol
m	Bottleneck dimension	Design parameter to be chosen
d_{ID}	Intrinsic dimension of the data manifold	Target quantity (unknown for real data)
d_{emb}	Minimum Euclidean embedding dimension of the specified manifold; for an ideal d -dimensional manifold, $d_{\text{emb}} \geq d$	Lower bound for m (Prop. 1)
$\hat{d}_{\text{ID}}$	TwoNN/MLE estimate of d_{ID}	Reference value for Step 1 (may be biased)
$\text{AU}(m)$	Number of KL-active coordinates, Equation (1)	Verification signal for Step 3
AU_{sat}	Saturated (plateau) AU value	Saturated number of empirically active coordinates under the specified β , δ , and architecture (VAE)
$d^*(\beta)$	Water-filling threshold count (Prop. 2)	Idealized prediction of AU_{sat}
κ	Decoder Jacobian condition number	Isometry (distortion) indicator

Two distinctions deserve emphasis here. First, $\hat{d}_{\text{ID}}$ is a **reference value, not the true value**: even when TwoNN/MLE is numerically stable (low variance), it may carry a systematic bias due to geometric distortion of the reference representation (stability vs. validity; Section 6.2). Second, AU is **not an estimator of the intrinsic dimension**: AU is an observable—“the number of coordinates the KL regularization allowed to be used”—and depends on β , δ , and the architecture. The protocol is designed to avoid over-reliance on any single indicator by cross-checking these two quantities of different natures.

3.2. The Protocol

Figure 1 summarizes the three steps; the following subsections specify each step.

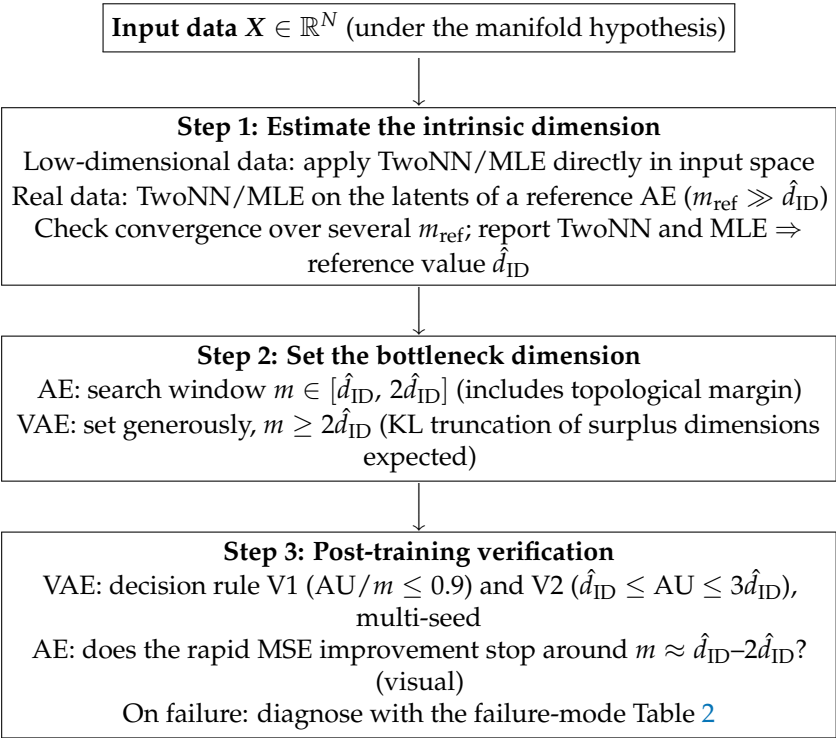


Figure 1. ID-guided protocol for choosing the bottleneck dimension m . The protocol closes the loop: the data-side estimate $\hat{d}_{\text{ID}}$ determines the design, and the model-side observable AU verifies the design after training.

3.2.1. Step 1: Estimating the Intrinsic Dimension

For synthetic or low-dimensional data, TwoNN/MLE can be applied directly in input space. For high-dimensional real data (e.g., $N = 784$ for MNIST), direct estimation becomes unstable due to distance concentration, so we estimate **through the latent representation of a generous reference AE** ($m_{\text{ref}} \gg \hat{d}_{\text{ID}}$; $m_{\text{ref}} = 256$ in this paper). Two requirements are mandatory: (i) confirm convergence of the estimate over several m_{ref} (e.g., $\{64, 128, 256\}$); (ii) report two estimators with different principles (TwoNN and MLE) and confirm they agree in order of magnitude. For setting the window and the verification point in Steps 2–3, the **pre-registered primary estimator is the converged TwoNN value rounded to the nearest integer**; MLE (and, where computed, DANCo/ESS) serve as cross-checks (how a diverging cross-check is handled is specified in Step 3). The reference representation used in Step 1 must be trained **independently of the candidate VAE models evaluated in Step 3**; latent-space estimates reported for candidate VAEs (e.g., Table 8) are diagnostic cross-checks and are never used to define $\hat{d}_{\text{ID}}$. That the condition $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ is the dominant condition for stable estimation is consistent with the conditional-stability rationale of Section 4.3 and supported by the experiments (Section 5.5). Conversely, $m_{\text{ref}} \lesssim 3\hat{d}_{\text{ID}}$ causes systematic underestimation, and on natural images inflation occurs for $m_{\text{ref}} \gg 2\hat{d}_{\text{ID}}$ (Section 5.6)—note that the trust region of Step 1 is bounded in m_{ref} .

3.2.2. Step 2: Setting the Bottleneck Dimension

For AEs: for $m < d_{\text{emb}}$, zero-error self-intersection-free reconstruction is impossible in principle under the idealized continuous-manifold assumption (Proposition 1), and any excess in m remains as redundant dimensions (an AE has no automatic truncation). We therefore take $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$ as the **default search window**, absorbing the possibility of a topological obstruction ($d_{\text{emb}} > d_{\text{ID}}$; Torus type) into the margin. Here we re-emphasize the distinction of Section 1.3: what the bottleneck actually requires is d_{emb} , but TwoNN/MLE can only estimate d_{ID} , and d_{emb} is unknown for real data. **The multiplier 2 at the upper end of the window acts as a buffer against this unknown gap $d_{\text{emb}} - d_{\text{ID}}$ (the topological obstruction)**— $\hat{d}_{\text{ID}}$ must not be misread as an estimate of d_{emb} . Choose the upper side of the window if reconstruction accuracy matters, the lower side if compression matters. The factor 2 is not derived from first principles but is a default that worked in our experiments; topology, noise, class mixtures, or reconstruction-fidelity demands may require a larger margin (on the Torus, activation stabilizes only at $m \geq 5 > 2\hat{d}_{\text{ID}} = 4$; Section 5.2)—the window is a starting point for the search, not a guaranteed upper bound.

For VAEs: Proposition 2 predicts, within the surrogate model, that the KL regularization truncates surplus dimensions even when m exceeds the required dimension, so **setting $m \geq 2\hat{d}_{\text{ID}}$ generously is the conservative operating strategy** (whether truncation actually occurred is checked by V1 in Step 3). For the VAE, too, the role of the multiplier 2 is the same as for the AE: a margin that keeps m from falling below the unknown $d_{\text{emb}} \geq d_{\text{ID}}$ (as long as truncation operates, any excess is reported post hoc by AU). The truncation threshold, however, depends on β ($d^*(\beta) = |\{j : \lambda_j > \beta\tau^2\}|$): if β is too small the truncation fails and AU sticks at m (detectable in Step 3).

3.2.3. Step 3: Post-Training Verification

After training, the design is verified with two signals (both over multiple seeds).

VAE: AU verification (with a pass/fail decision rule). The verification point m_{ver} is **pre-specified as the smallest tested grid point $\geq 2\hat{d}_{\text{ID}}$ —with $\hat{d}_{\text{ID}}$ the primary TwoNN estimate of Step 1—before the AU curve is observed** (to avoid post-hoc selection). When a cross-check estimator (e.g., MLE) yields a $2\hat{d}_{\text{ID}}$ that falls past the next grid point, verify there as well (on MNIST, MLE ≈ 11.9 points to $m = 32$ in addition to $m = 20$, and both

pass; Section 5.3). For a VAE trained at m_{ver} , evaluate the following two conditions on the multi-seed mean AU:

$$\text{V1 (truncation occurred)} : \text{AU}(m_{\text{ver}})/m_{\text{ver}} \leq 0.9, \quad (2)$$

$$\text{V2 (order consistency)} : \hat{d}_{\text{ID}} \leq \text{AU}(m_{\text{ver}}) \leq 3\hat{d}_{\text{ID}}. \quad (3)$$

The verification **passes** when both hold (evidence that AU lies within the operational range $[\hat{d}_{\text{ID}}, 3\hat{d}_{\text{ID}}]$ defined relative to the Step-1 reference); failure of V1 suggests “ β too small relative to the decoder capacity,” failure of the upper side of V2 suggests “Step-1 underestimation or insufficient truncation,” and failure of the lower side suggests “over-regularization (β too large)” (Table 2). V1 and V2 are **operational screening criteria introduced for this study**: they are not hypothesis tests, confidence rules, or universally calibrated acceptance criteria, and the threshold 0.9 and factor 3 are working defaults. In all settings of this paper, the verdicts on the multi-seed mean AU are invariant for a V1 threshold in (0.67, 0.94) and a V2 factor in (2.3, 6.8); recomputed per seed, all per-seed verdicts agree at the default thresholds and are invariant for a V1 threshold in (0.70, 0.93) and a V2 factor in (2.3, 6.6)—the narrower per-seed V1 lower edge is set by one MNIST seed with $\text{AU} = 14$ at $m_{\text{ver}} = 20$. The verdicts concerning the lower V2 bound (default coefficient 1) are likewise invariant for a lower coefficient in (0.70, 1.0], whose lower edge is set by the Fashion-MNIST boundary case $\text{AU}/\hat{d}_{\text{ID}} = 0.70$ (Section 5.3.6). (Section 5; the supplementary analysis script recomputes all ranges; the displayed V2 ranges are rounded conservatively—the computed mean-verdict lower edge is 2.23.) The width of the V2 band $[\hat{d}_{\text{ID}}, 3\hat{d}_{\text{ID}}]$ is a consequence of AU being a β - and architecture-dependent observable rather than an ID estimator (Section 3.1); V2 checks order consistency only, not that AU equals any dimension quantity—in particular, a pass does not imply $\text{AU}_{\text{sat}} = d_{\text{emb}}$ (see the Torus boundary behavior in Section 5.2). More generally, a V1/V2 pass indicates only that the observed AU is compressed and lies within the operational range defined relative to the Step-1 reference under the specified conditions. It does not establish that every surplus coordinate has fully collapsed, does not establish equality between AU and the intrinsic or embedding dimension, and does not guarantee reconstruction quality, generative quality, or downstream utility; the rule is a diagnostic to be read together with MSE (and, where relevant, ELBO, downstream metrics, and visualization). Simple manifolds show sharp saturation, while real data show a gradual slowdown (Section 5.3).

MSE verification (qualitative diagnostic; AE and VAE). Visually confirm that the rapid improvement of $\text{MSE}(m)$ slows down around $m \approx \hat{d}_{\text{ID}} - 2\hat{d}_{\text{ID}}$ (we do not propose an algorithmic elbow detector; automatic change-point detection is strongly grid- and seed-dependent and is confined to the exploratory analysis of Appendix B). For VAEs the MSE verification must **always be read together with V1/V2**: if AU passes but MSE keeps improving markedly beyond m_{ver} , suspect over-pruning due to a topological obstruction ($d_{\text{emb}} > d_{\text{ID}}$) and enlarge the margin (fifth row of Table 2; a concrete instance on the Torus is given in Section 5.2). The protocol is thus a **diagnostic workflow** that combines the mechanical V1/V2 verdict with the required plots and statistics (see the reporting requirements below); it is not a selection algorithm that outputs a single number.

Failure modes for a mismatched verification are organized in Table 2; this is a diagnostic checklist for readers applying the protocol to new data or architectures.

Table 2. Failure modes in Step 3 and their diagnoses. Each row is backed by the experiments in Section 5.

Symptom	Diagnostic interpretation	Remedy / follow-up
$AU = m$ for all tested m (no truncation)	Consistent with β being too small relative to decoder capacity, or with the effective truncation threshold lying beyond the tested range	Increase β or restrict decoder capacity; see Conv-VAE boundary (Section 5.6)
AU saturates far above $\hat{d}_{ID}$	Step-1 underestimate (m_{ref} too small), or entangled multi-class structure	Re-check Step-1 convergence in m_{ref} ; treat AU_{sat} as an upper reference
AU saturates below $\hat{d}_{ID}$ (lower V2 bound violated)	Consistent with over-regularization (β too large for the data/decoder) or with the Step-1 reference containing directions the VAE prunes at this β	Sweep β ; cross-read the MSE and downstream metrics; treat the interval $[AU, \hat{d}_{ID}]$ as a diagnostic range (Fashion-MNIST; Section 5.3.6)
$\hat{d}_{ID}$ inflates as m_{ref} grows (no convergence)	Latent-geometry distortion at large m_{ref} (natural images)	Restrict to $m_{ref} \approx 2\hat{d}_{ID}$ and cross-check with the input-space estimate
MSE keeps improving beyond $2\hat{d}_{ID}$	Topological obstruction ($d_{emb} > d_{ID}$) or noise fitting	Allow a larger margin; inspect reconstruction quality

Reporting requirements: when reporting an application of this protocol, state at minimum (i) the number of seeds and the inter-seed variance of $\hat{d}_{ID}$ and AU, (ii) the m -grid and the values of m_{ref} , (iii) β , δ , and preprocessing, and (iv) both the TwoNN and MLE estimates. The main experiments of this paper report these quantities where applicable; for the boundary and exploratory experiments the complete set (including MLE and inter-seed variances) is contained in the supplementary logs.

3.3. Applicability (Summary in Advance)

Table 3 summarizes in advance where the protocol works and where it reaches its boundary (evidence in Section 5). Readers can first check whether their setting is within the applicable range.

Table 3. Applicability of the ID-guided protocol (summary of the evidence in Section 5). “Stable” means low inter-seed variance under our experimental conditions, not unbiasedness.

Setting	Step 1 ($\hat{d}_{\text{ID}}$)	Step 3 (AU verification)	Overall
Synthetic folds ($d_{\text{true}} \leq 3$, 3–5 seeds)	Direct input-space estimate; matches d_{true}	Pass: saturation at $d_{\text{true}} / d_{\text{emb}}$ with sufficient margin ($\sigma = 0$); Torus boundary over-pruning ($m \leq 4$) caught by the MSE cross-check	Validated on synthetic ground truth when adaptive margin enlargement is allowed (triggered by the MSE cross-check)
FC-VAE on MNIST / Fashion-MNIST (3 seeds each)	Stable: ≈ 10 (ref.-AE sweep; VAE-latent cross-check 10.03 ± 0.12) / ≈ 10	Pass (V1/V2) on MNIST; Fashion-MNIST: V1 pass, V2 lower-bound violation (over-pruning boundary diagnosed)	Applicable (MNIST fully validated; on Fashion-MNIST the rule detects a boundary case)
Conv-VAE on MNIST	Stable (inflates while $\text{AU} = m$)	Fail at $\beta_{\text{max}} = 4$ (V1); $\beta_{\text{max}} \geq 10$ restores V1 but not V2 (3 seeds)	Conditional applicability after architecture- and schedule-dependent truncation is induced; order consistency remains open
Natural images (CIFAR-10 / SVHN, 3 seeds)	CIFAR-10 consistent with Pope et al. [6] at $m_{\text{ref}} \approx 2\hat{d}_{\text{ID}}$; inflates beyond	Unresolved: no V1/V2 pass within tested range	Step 1 usable with care; Step 3 open problem

4. Theoretical Motivation and Interpretive Surrogates

We state the status of the results up front. Proposition 1 is a rigorous lower bound under an idealization (zero reconstruction error); Proposition 2 is an analytical result **restricted to the linear-Gaussian setting**, and its application to nonlinear VAEs is a qualitative guide (an interpretive surrogate); Section 4.3 gives a **heuristic conditional-stability rationale** (not a finite-sample theorem) that grounds the design of Step 1.

4.1. Necessary Condition: Topological Lower Bound

Proposition 1 (Topological lower bound for the bottleneck dimension). *Let $\mathcal{M} \subset \mathbb{R}^N$ be a compact smooth d -dimensional manifold and let $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$ be continuous. If $g \circ f$ is the identity on $\mathcal{M}$ (zero reconstruction error), then $m \geq d_{\text{emb}} \geq d$, where d_{emb} is the topological minimum embedding dimension of $\mathcal{M}$.*

Proof. From $g(f(x)) = x$ for $x \in \mathcal{M}$, the restriction $f|_{\mathcal{M}}$ is a continuous injection. A continuous injection from a compact space into a Hausdorff space is a homeomorphism onto its image, so $f|_{\mathcal{M}}$ is a topological embedding; no topological embedding of a d -dimensional manifold into $\mathbb{R}^m$ with $m < d_{\text{emb}}$ exists by the definition of d_{emb} [22]. $\square$

The proposition provides a necessary condition for zero-error, self-intersection-free reconstruction **under an idealized continuous-manifold assumption**; it does not directly apply to real neural AEs with finite samples, finite reconstruction error, noise, or non-injective behavior on small sets, and we use it only as an approximate guide for the lower end. This is the rationale for the lower end of Step 2. For real data d_{emb} is unknown; for an ideal d -dimensional smooth manifold $d_{\text{emb}} \geq d$ holds, but $\hat{d}_{\text{ID}}$ estimates local intrinsic dimensionality rather than d_{emb} , so the relation is a geometric motivation for adding a margin—not an empirically verifiable inequality involving $\hat{d}_{\text{ID}}$ (Section 1.3). In this spirit, $\hat{d}_{\text{ID}}$ provides a practical reference point below which insufficient representation capacity is

more likely under the tested conditions—it is not a formal lower bound for real datasets—and the possibility $d_{\text{emb}} > d_{\text{ID}}$ (a Torus-type topological obstruction) is one reason for the margin “set m above $\hat{d}_{\text{ID}}$.” The synthetic experiments provide instances in which this idealized lower bound is empirically echoed as the MSE elbow for the Swiss Roll ($d_{\text{emb}} = 2$) and the Torus ($d_{\text{emb}} = 3 > d_{\text{ID}} = 2$) (Section 5.2); we regard these as illustrations, not as a direct empirical proof of the proposition for finite-error networks.

4.2. Mechanism on the Sufficient Side: Water-Filling Surrogate Activation Condition

Proposition 2 (Surrogate activation condition under a linear-Gaussian rate–distortion approximation). Consider a β -VAE with linear encoder/decoder and Gaussian prior (details in Appendix A). Here τ^2 is the **isotropic Gaussian observation-noise variance of the decoder output** (the variance parameter of the likelihood $p(x|z) = \mathcal{N}(g(z), \tau^2 I)$), and λ_j are the **eigenvalues of the covariance matrix of the centered input data**. We further use the rate–distortion approximation that replaces the KL term by mutual information (this replacement is an **approximation** that neglects the mismatch between the aggregated posterior and the prior; the validity of this proposition is limited to the range of this approximation). Then, from the water-filling theorem and the KKT conditions:

1. (Activation threshold) The j -th eigendirection is active if and only if

$$\lambda_j > \beta \tau^2. \quad (4)$$

2. (Optimal number of active directions) The number of active eigendirections at the global optimum is $A^*(m) = \min(m, d^*(\beta))$ with $d^*(\beta) := |\{j : \lambda_j > \beta \tau^2\}|$.
3. (Monotone saturation) $A^*(m)$ is nondecreasing in m and saturates for $m \geq d^*(\beta)$.

This proposition is not a theorem about trained nonlinear VAEs but an **interpretive surrogate** for reading nonlinear experiments (derivation in Appendix A). The activation condition (4) is exact only for the specified linear-Gaussian surrogate; in nonlinear VAEs it must not be interpreted as a coordinate-wise activation theorem or as a quantitative predictor of the empirical AU—it only motivates the qualitative expectation that increasing β , or reducing the effective decoder capacity, can promote latent truncation.

Remark 1 (Implications for the protocol). All statements below are **predictions within the surrogate model** and are operated as diagnostic expectations for nonlinear VAEs (their success or failure is judged in Step 3). (i) Surplus dimensions with $m > d^*(\beta)$ are truncated at the optimum of the surrogate—this is the rationale for Step 2, “a generous m is safe for VAEs.” (ii) The truncation threshold $\beta \tau^2$ is proportional to β —if β is too small then $d^*(\beta) > m$ and no truncation occurs ($\text{AU} = m$). This is the failure mode in the first row of Table 2 and is actually observed for Conv-VAEs and natural images (Section 5.6). (iii) For $\beta = 0$ (vanilla AE) the threshold vanishes and all dimensions are used—the reason an AE has no automatic truncation and m itself must be constrained in Step 2.

Remark 2 (Surrogate rate vs. empirical AU). What Proposition 2 counts is the number of eigendirections with positive rate $R_j^* > 0$; what the empirical AU (Equation (1)) counts is the number of coordinates whose posterior-mean variance exceeds a threshold—by definition, two different notions. In the linear-Gaussian setting they coincide: at the optimum, $R_j^* > 0 \iff$ the posterior mean $\mu_j(x)$ depends on x ($\text{Var}_x[\mu_j] > 0$) $\iff$ AU counts the coordinate as active, while $R_j^* = 0 \iff \mu_j \equiv 0$ (collapse). For nonlinear VAEs this correspondence is only a **qualitative expectation**, and we do not claim that the surrogate predicts nonlinear AU counts quantitatively. In actual nonlinear VAEs, each m is optimized independently and stochastically, so the empirical $\text{AU}(m)$ carries non-monotone fluctuations, and the sharp saturation of the linear setting is replaced by a gradual slowdown (Section 5.3). Moreover, replacing the effective noise variance of a nonlinear decoder by a calibrated value τ_{eff}^2 “back-computed to match the observed effective dimension”

(Section 5.6) is a **heuristic extrapolation**, not a consequence of this proposition. We note that the mismatch term $\text{KL}(q_\phi(z) \parallel p(z))$ dropped by the rate–distortion approximation vanishes on the truncated coordinates of the globally optimized linear-Gaussian solution and is assumed negligible on its active coordinates (Appendix A), and during nonlinear optimization it is nonnegative and can grow as the aggregated posterior develops cluster structure—from this sign, the surrogate tends to **predict weaker truncation** than would be induced by the additional aggregated-posterior mismatch; equivalently, it may overestimate the number of active dimensions relative to the nonlinear model (details in Appendix A).

4.3. Grounding of Step 1: A Conditional-Stability Rationale for Latent-Space TwoNN Estimation

This subsection provides a **heuristic conditional-stability rationale rather than a finite-sample theorem** for TwoNN; we deliberately do not state it as a numbered proposition, because no explicit constants, finite-sample error bounds, or estimator-continuity conditions are derived.

Conditional-stability rationale (heuristic): if the encoder f is locally bi-Lipschitz (distortion $\varepsilon \in [0, 1)$) and neighbor ranks are preserved with high probability (Trustworthiness $\geq 1 - \delta_T$ with small δ_T), then the change in the distribution of the two-neighbor distance ratio r_2/r_1 is expected to be bounded by a continuous function $C(\varepsilon, \delta_T)$ (with $C \rightarrow 0$ as $\varepsilon, \delta_T \rightarrow 0$), and the deviation between the input-space and latent-space TwoNN estimates is governed by the same bound. This argument is not a guarantee of unbiasedness but a heuristic description of conditions that may support conditional stability (supporting argument in Appendix A).

The experiments (Section 5.5) show that improving δ_T (neighbor-rank preservation) via $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ dominates over improving ε (isometry)—this is the design rationale for prioritizing bottleneck margin over isometric regularization in Step 1.

5. Experiments

The experiments are organized in three layers: (i) **principle validation**—does the protocol return the correct dimension on synthetic manifolds with known ground truth (Section 5.2)? (ii) **real-data validation**—does the full protocol work on MNIST, what evidence is available for Fashion-MNIST, and does it have practical value in search cost and downstream performance (Sections 5.3–5.5)? (iii) **applicability boundary**—which conditions break for Conv-VAEs and natural images (Section 5.6)? **All main-text VAE verification results used for the principal verdicts are multi-seed experiments with at least 3 seeds (42, 123, 777)**; two reused training sweeps are single-seed and are labeled as such where they appear (the Step-1 reference-AE sweep of Table 6, also re-analyzed in Tables 7 and 11, and the AE side of the downstream comparison); single-seed exploratory observations (the natural-image AU-plateau search and the topological-manifold AU_{sat} observations) are labeled as such where summarized, with details in Appendix C.

5.1. Experimental Setup

Common: Adam ($\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, no weight decay), batch size 128, ReLU (Sigmoid output), PyTorch 2.0, a single NVIDIA GeForce RTX 3070 GPU (CPU fallback when CUDA is unavailable). Preprocessing is $[0, 1]$ pixel normalization only (synthetic manifolds are standardized to zero mean and unit variance). Train/test subsets use the first $n_{\text{train}}/n_{\text{test}}$ samples of the official splits in fixed order. We intentionally use these fixed subsets, rather than the complete official training sets, to control computational cost and to keep conditions identical across experiments, and we take the first samples to ensure exact reproducibility; because this choice may introduce order-specific effects, the reported results should not be interpreted as estimates averaged over independent

data partitions. AU, TwoNN, and MLE are computed on the test subset. Unless otherwise stated, all $\pm$ values in multi-seed experiments denote standard deviations across seeds, not standard errors or confidence intervals.

Synthetic manifolds: hidden layers [64, 32], 200–300 epochs, VAE $\beta = 4.0$, $n = 5,000$ (4,000 train / 1,000 test).

MNIST [23] / Fashion-MNIST [24]: input dimension 784, hidden layers [512, 256], VAE $\beta = 4.0$, m -grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$ (changes per experiment are stated); MNIST: 300 epochs ($n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$); Fashion-MNIST: 300 epochs (Exp-Real-Geom, Exp-Fashion-MS; $n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$).

ID estimation: TwoNN ($k = 2$) [5], MLE ($k = 10$; stability confirmed for $k \in \{5, 10, 20\}$) [13]. AU threshold $\delta = 10^{-2}$ (Appendix D).

Reproducibility and supplementary-file inventory: the supplementary material consists of four parts: (i) the experiment scripts (in one-to-one correspondence with the experiment names in Table A4); (ii) analysis scripts that regenerate every table and figure of the main text from the training logs; (iii) the seed-specific training logs (JSON; all seeds); and (iv) configuration files, data-subset indices, and environment information (random seeds, library and framework versions). We verified before submission that the numbers in the tables match the output of the included scripts. The material is attached for review; upon acceptance it will be deposited in a public repository (GitHub/Zenodo) with URL and DOI.

5.2. Principle Validation: Synthetic Manifolds with Known Ground Truth

We validated the three protocol steps against ground truth on the Swiss Roll ($d_{\text{true}} = d_{\text{emb}} = 2$) and the Torus ($d_{\text{true}} = 2$, $d_{\text{emb}} = 3$) (3 seeds).

Step 1: the input-space TwoNN gives ≈ 2.0 for both the Swiss Roll and the Torus, matching d_{true} .

Step 3 (AE: MSE verification): as shown in Table 4, the rapid MSE improvement stops at $m = 2 = d_{\text{emb}}$ for the Swiss Roll and at $m = 3 = d_{\text{emb}}$ for the Torus—the lower bound of Proposition 1 appears as the actual MSE elbow, and for the Torus the topological obstruction “ $d_{\text{ID}} = 2$ is not enough” is observed. This is the experimental basis for the margin in Step 2 (set m above $\hat{d}_{\text{ID}}$).

Table 4. Synthetic manifolds (Exp-Syn-1): AE reconstruction MSE (mean $\pm$ std, 3 seeds). The elbow appears exactly at $m = d_{\text{emb}}$.

m	Swiss Roll MSE	Torus MSE
1	1.016 ± 0.315	0.356 ± 0.022
2	$0.029 \pm 0.001 \leftarrow \text{elbow} (= d_{\text{emb}})$	0.109 ± 0.007
3	$(6 \pm 2) \times 10^{-4}$	$(4 \pm 3) \times 10^{-4} \leftarrow \text{elbow} (= d_{\text{emb}})$
4	$(5 \pm 1) \times 10^{-4}$	$(1.0 \pm 0.4) \times 10^{-4}$

Step 3 (VAE: AU verification): as shown in Table 5, the AU of the VAE ($\beta = 4$) saturates at $\text{AU}_{\text{sat}} = 2 = d_{\text{true}}$ for the Swiss Roll over the whole tested range $m \in [2, 16]$, and at $\text{AU}_{\text{sat}} = 3 = d_{\text{emb}}$ for the Torus for $m \geq 5$, in all seeds (standard deviation 0.0; the activation at $m = 5$ is from the fine-grid experiment Exp-Syn-Verify). In contrast, the coordinate usage of the vanilla AE sticks to m (no truncation mechanism)—consistent with implications (i) and (iii) of Proposition 2: the surrogate prediction that *the VAE truncates the surplus dimensions of a generous m and AU reports the result* is confirmed on ground truth in this setting. However, **in the boundary zone $m \in \{3, 4\}$ the Torus is over-pruned to $\text{AU} = 2 < d_{\text{emb}}$** —detailed next.

Table 5. Synthetic manifolds (Exp-Syn-AU): active units of the VAE ($\beta = 4$) vs. the coordinate usage of the vanilla AE (mean $\pm$ std, 3 seeds). “AU” for the AE is a pseudo-AU defined by the encoding-variance threshold (auxiliary observation). On the Swiss Roll the VAE AU saturates at $2 = d_{\text{true}}$ over the whole tested range; on the Torus, $\text{AU} = 2$ at $m = 4$ is an over-pruned boundary case ($< d_{\text{emb}}$), and $\text{AU} = 3 = d_{\text{emb}}$ becomes stable from $m = 5$ in the fine-grid experiment (see text).

	m	Vanilla AE	VAE ($\beta = 4$)
Swiss Roll ($d_{\text{true}} = d_{\text{emb}} = 2$)	2	2	2.0 ± 0.0
	4	4	2.0 ± 0.0
	8	8	2.0 ± 0.0
	16	16	2.0 ± 0.0
Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$)	4	4	2.0 ± 0.0
	6	6	3.0 ± 0.0
	8	8	3.0 ± 0.0
	16	16	3.0 ± 0.0

Boundary-zone over-pruning and its detection by the MSE cross-check (Exp-Syn-Verify): for $m \in \{3, 4\}$ the Torus VAE settles at $\text{AU} = 2 < d_{\text{emb}} = 3$. This is neither a reporting error nor a fluke but **a genuine phenomenon that reproduces across seeds:** in a fine-grid confirmation experiment ($m \in \{3, 4, 5, 6, 8\}$, 5 seeds; Exp-Syn-Verify), all seeds give $\text{AU} = 2$ for $m \in \{3, 4\}$ and $\text{AU} = 3$ for $m \geq 5$ —the transition at $m = 5$ was identical across all five tested seeds under this training setup. Over-pruning carries a clear reconstruction penalty: the MSE improves rapidly from 0.220 ± 0.002 at $m = 4$ to 0.138 ± 0.001 at $m = 5$ (where the third coordinate activates) and 0.073 ± 0.006 at $m = 6$, because a self-intersection-free embedding becomes possible. The interpretation is that under the KL pressure of $\beta = 4$, in the boundary zone with barely any surplus dimensions the reconstruction gain of keeping a third coordinate does not overcome its KL cost, and the model is consistent with a two-dimensional representation that cannot provide a globally self-intersection-free embedding of the torus (geometrically, this may correspond to an immersion-like or self-overlapping representation, although we did not directly visualize self-intersections)—**the activation of the topologically required dimension stabilizes across all seeds only for $m \geq 5$, just outside the upper end $2\hat{d}_{\text{ID}} = 4$ of the default search window.** This observation has two operational implications. First, it demonstrates on ground truth why the Step-2 margin is needed: on the Torus, the topological gap $d_{\text{emb}} > d_{\text{ID}}$ makes the default window $[\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}] = [2, 4]$ insufficient, and stability is reached only outside the window—the reason the window is a starting point, not a guaranteed upper bound (Section 3.2.2). Second, it **makes a limitation of V1/V2 explicit:** at the pre-specified verification point $m_{\text{ver}} = 4 = 2\hat{d}_{\text{ID}}$, $\text{AU} = 2$ passes both V1 ($r = 0.5$) and V2 ($2 \leq 2 \leq 6$)—because V1/V2 verifies order consistency with $\hat{d}_{\text{ID}}$, not that $\text{AU}_{\text{sat}} = d_{\text{emb}}$. What catches this miss is the accompanying MSE verification: the MSE keeps improving markedly beyond $m_{\text{ver}} = 4$ ($0.22 \rightarrow 0.07$), so the fifth row of Table 2 (topological obstruction) fires and an enlarged margin is prescribed.

Interpretive caution (distinguishing d_{ID} , d_{emb} , and AU): this result must not be read as “AU estimates d_{emb} .” What TwoNN/MLE estimate is the local intrinsic dimension d_{ID} ; what appears in the lower bound of Proposition 1 is the topological embedding dimension d_{emb} ; and AU is an observable “number of used coordinates” that depends on β , δ , and the architecture (Table 1)—three quantities with different definitions. The correspondence $\text{AU}_{\text{sat}} = d_{\text{emb}}$ on the Torus is a **conditional empirical observation**—under sufficient β , convergence, standardization preprocessing, and sufficient margin ($m \geq 5$ on the Torus), the number of coordinates remaining after KL truncation happened to coincide with the dimension required for embedding—not a theoretical necessity (generalization beyond the

Torus and Möbius band is unconfirmed; single-seed topological-manifold experiments are in Appendix C).

Verdict (Step-3 rule; m_{ver} pre-specified as the smallest tested point $\geq 2\hat{d}_{\text{ID}}$): the Swiss Roll ($m_{\text{ver}} = 4 = 2\hat{d}_{\text{ID}}$; AU = 2.0, activation ratio 0.50) **passes** V1, V2, and the MSE verification (3 seeds, $\sigma = 0$). The Torus at $m_{\text{ver}} = 4$ passes V1 and V2, but **the MSE verification detects the topological obstruction** (a $3\times$ MSE improvement from $m = 4$ to $m = 6$, $0.220 \rightarrow 0.073$)—following the fifth row of Table 2, the margin is enlarged and all checks pass at $m = 6$ (**pass**; $\text{AU}_{\text{sat}} = 3 = d_{\text{emb}}$, stable for all larger m ; 3–5 seeds, $\sigma = 0$). This two-stage trajectory is itself a working example of the diagnostic workflow in which AU is always read together with MSE.

5.3. Real-Data Validation: MNIST

5.3.1. Step 1: Estimating $\hat{d}_{\text{ID}}$

We computed TwoNN/MLE on the latents of reference AEs (300 epochs; Table 6). TwoNN converges to around 10.0 for $m_{\text{ref}} \geq 64$, and MLE (11.3–11.9) gives the same order. (As a diagnostic cross-check—**not** used to define $\hat{d}_{\text{ID}}$ —the 3-seed TwoNN value on the latents of the Step-3 candidate VAEs is 10.03 ± 0.12 ; Table 8.) The Step-1 requirements—convergence over several m_{ref} and agreement of the two estimators—are met: the pre-specified rounded-TwoNN rule yields the **operational primary reference** $\hat{d}_{\text{ID}} = 10$ for **MNIST**, while the MLE values of 11.3–11.9 provide an order-of-magnitude cross-check (Section 3.2.1). Note that the m_{ref} -convergence sweep itself was performed with a single seed; estimator uncertainty was assessed by subsampling (Section 5.3.2), and the latent-space estimate was reproduced across three VAE seeds, but a full multi-seed reference-AE sweep over all m_{ref} values was not performed (Section 6.4). Note that each row of Table 6 is itself an m -sweep of a plain AE, so its MSE column is also used for the Step-3 AE-side MSE verification below. As a contrast, linear PCA requires 43 dimensions for 80% cumulative variance and 86 for 90% (Figure 2, left)—cumulative variance dimensions are not intrinsic-dimension estimates, so this is an illustrative contrast rather than a direct baseline, but it shows that the dimensionality required for linear compression far exceeds the nonlinear estimate $\hat{d}_{\text{ID}} \approx 10$ –12.

Table 6. MNIST Step 1 (reference-AE latents used to define $\hat{d}_{\text{ID}}$): TwoNN/MLE estimates through reference AEs of increasing m_{ref} . **All rows are a single-seed (42) exploratory sweep** (300 epochs); statistical uncertainty at $m_{\text{ref}} \in \{64, 128, 256\}$ is quantified separately by subsampling CIs and estimator cross-checks in Table 7 (Section 5.3.2), and the three-seed candidate-VAE latent cross-check (diagnostic only; not used to set $\hat{d}_{\text{ID}}$) is reported in Table 8.

m_{ref}	MSE	TwoNN ID	MLE ID	CKA [25]
8	0.0191	6.08	6.70	0.725
16	0.0121	7.96	8.61	0.812
32	0.0087	8.84	9.89	0.875
64	0.0062	9.54	11.26	0.908
128	0.0061	9.84	11.45	0.921
256	0.0057	10.02	11.85	0.953

5.3.2. Statistical Robustness of Step 1: Subsampling CIs and Estimator Cross-Checks

Since $\hat{d}_{\text{ID}}$ provides the center of the search window, we quantify its statistical uncertainty (Exp-Boot-CI; a re-analysis of the latents of the trained reference AEs). Because bootstrap resampling with replacement corrupts kNN-based estimators through duplicated points, we report 95% percentile intervals from subsampling without replacement ($n = 1500$, 200 repeats). In addition, we apply DANCo [14] and ESS [26] (scikit-dimension

implementations)—estimators of different principles—to the same latents and check order-of-magnitude agreement with TwoNN/MLE.

Table 7. Exp-Boot-CI (MNIST): statistical robustness of Step 1. Subsampling 95% intervals ($n = 1500$, 200 repeats, without replacement) and cross-check estimators (DANCo, ESS) on the same reference-AE latents (single-seed-42 models; the uncertainty quantified here is the sampling variability of the estimators).

Representation	TwoNN [95% CI]	MLE [95% CI]	DANCo	ESS
Raw input ($N = 784$)	12.63	14.46	784 (degenerate)	18.67
AE latents, $m_{\text{ref}} = 64$	9.56 [8.92, 10.24]	11.14 [10.91, 11.41]	10.04	13.17
AE latents, $m_{\text{ref}} = 128$	9.72 [9.22, 10.23]	11.33 [11.08, 11.56]	10.03	13.72
AE latents, $m_{\text{ref}} = 256$	9.93 [9.37, 10.68]	11.67 [11.45, 11.90]	10.95	14.98

The results (Table 7) can be summarized in three points. First, **the subsampling 95% intervals are narrow**—width 1.0–1.3 for TwoNN and 0.5 for MLE—so the read-out of the search-window center $\hat{d}_{\text{ID}} \approx 10$ –12 is robust to sampling variability. These intervals quantify subsampling variability **conditional on the trained reference representation**; they do not include variation due to model initialization, training stochasticity, dataset resampling at the population level, or estimator bias. Second, **DANCo, an estimator of a different principle, agrees closely with TwoNN** at 10.0–10.9, and ESS is somewhat higher at 13–15 but of the same order—the Step-1 read-out is not an artifact peculiar to TwoNN behavior. In the raw input space, TwoNN (12.63) and MLE (14.46) deviate only moderately from the latent estimates, whereas DANCo degenerates to the ambient dimension 784 (a known failure mode of its angle/norm-statistics calibration when the ambient dimension is large relative to the sample size, not a software fault) and ESS inflates to 18.7. The reference-AE route is therefore **a pragmatic route that stabilizes the chosen estimators under our conditions**, not a uniquely validated one—some estimators give the same order even in raw input space. Third, the subsample-size sensitivity ($n = 500$ –2500; see the supplementary material) shows the TwoNN mean rising monotonically with n (e.g., $9.07 \rightarrow 9.72$ at $m_{\text{ref}} = 64$)—a direct illustration of this paper’s characterization of TwoNN as “numerically stable but with a sample-size-dependent bias” (Section 6.2); the subsample means sit slightly below the full-sample values of Table 6 for this reason.

5.3.3. Step 2: Setting the Search Window

Using the rounded primary TwoNN reference $\hat{d}_{\text{ID}} = 10$, the default AE window of Section 3.2.2 would be $[10, 20]$; in the practical comparison reported here we use the broader window $m \in [10, 24]$, whose lower end is the rounded TwoNN estimate ($m_{\text{min}} = \text{round}(\hat{d}_{\text{TwoNN}}) = 10$) and whose upper end incorporates the MLE cross-check ($m_{\text{max}} = \lceil 2\hat{d}_{\text{MLE}} \rceil = \lceil 2 \times 11.85 \rceil = 24$). This widening is a practical window construction for the MNIST grid used here, not part of the general protocol. For the VAE we use a generous $m \geq 20$ (the VAE verification point follows the primary-TwoNN rule of Section 3.2.3).

5.3.4. Step 3: Verification by AU and MSE (3 Seeds)

Table 8 shows $\text{AU}(m)$ of the VAE ($\beta = 4$, 300 epochs, 3 seeds). AU is highly reproducible across seeds ($\sigma \leq 0.5$; Figure 3); nearly all dimensions are active for $m \leq 12$ ($\text{AU} \approx m$), while the growth clearly slows down beyond $m > 12$, reaching only $\text{AU} = 22.3$ at $m = 64$ —the verification signal expected by Step 3: **KL truncation sets in near $\hat{d}_{\text{ID}} \approx 10$ –12**. On the AE side, the rapid MSE improvement also slows at $m \approx 16$ –20 (the MSE column of Table 6, which is the plain-AE m -sweep), consistent with the search window $[10, 24]$.

Verdict (Step-3 rule; $m_{\text{ver}} = 20$ pre-specified as the smallest tested grid point $\geq 2\hat{d}_{\text{ID}} = 20$ from the primary TwoNN estimate $\hat{d}_{\text{ID}} = 10$): $\text{AU} = 13.3 \pm 0.5$ with activation ratio

$0.67 \leq 0.9$ (V1 holds) and $\hat{d}_{\text{ID}} = 10 \leq 13.3 \leq 30 = 3\hat{d}_{\text{ID}}$ (V2 holds): **pass** (per-seed verdicts agree 3/3). The verdict also passes at the MLE-shifted verification point $m = 32$ (MLE ≈ 11.9 gives $2\hat{d}_{\text{ID}} \approx 24$, whose next grid point is 32; Section 3.2.3) and at $m = 64$ (activation ratios 0.49 and 0.35; AU = 15.7 and 22.3). The accompanying MSE verification also shows no sign of a topological obstruction: the rapid improvement slows at $m \approx 16$ –20.

Table 8. MNIST Step 3 (Exp-Mnist-Multi-300): AU and TwoNN ID of the VAE ($\beta = 4$, 300 epochs, mean $\pm$ std over 3 seeds). The AU growth slows beyond $m \approx \hat{d}_{\text{ID}}$, and the latent TwoNN converges to $\hat{d}_{\text{ID}} \approx 10$. The TwoNN column is computed on **candidate-VAE latents** and serves as a diagnostic cross-check only; it is not used to set $\hat{d}_{\text{ID}}$ (Section 3.2.1).

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 ± 0.0	3.98 ± 0.14	all dims active
8	8.0 ± 0.0	6.67 ± 0.15	
12	10.0 ± 0.0	7.41 ± 0.06	slowdown begins near $\hat{d}_{\text{ID}}$
16	11.7 ± 0.5	7.95 ± 0.21	
20	13.3 ± 0.5	8.37 ± 0.14	
32	15.7 ± 0.5	9.07 ± 0.10	
64	22.3 ± 0.5	10.03 ± 0.12	

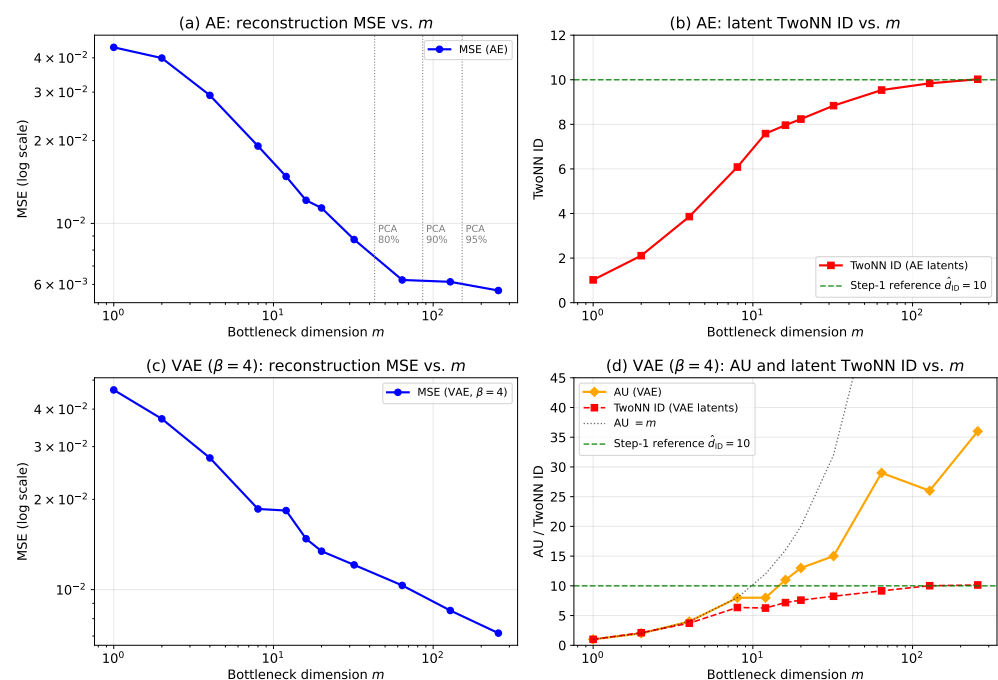


Figure 2. MNIST: m -dependence of the indicators (300 epochs; single-seed (42) sweep for both the AE and VAE panels—the multi-seed reproducibility of the VAE AU signal is shown separately in Figure 3). The single-seed VAE sweep is included for visualizing the dependence on m ; all principal VAE verification decisions are based on the independent three-seed results in Table 8 and Figure 3. **(a)** AE reconstruction MSE (log scale); vertical dotted lines mark the PCA 80/90/95% cumulative-variance dimensions (43/86/152), far above the nonlinear estimate $\hat{d}_{\text{ID}} \approx 10$. **(b)** TwoNN ID on the AE latents. **(c)** VAE ($\beta = 4$) reconstruction MSE. **(d)** VAE AU and latent TwoNN ID; the AU growth slows beyond $m \approx \hat{d}_{\text{ID}}$ (the vertical axis is clipped at 45 for readability, so the dotted AU = m diagonal leaves the plotted range). In panels (b) and (d), the dashed horizontal line marks the rounded Step-1 TwoNN reference $\hat{d}_{\text{ID}} = 10$ —an estimator-derived reference value, not a ground-truth dimension.

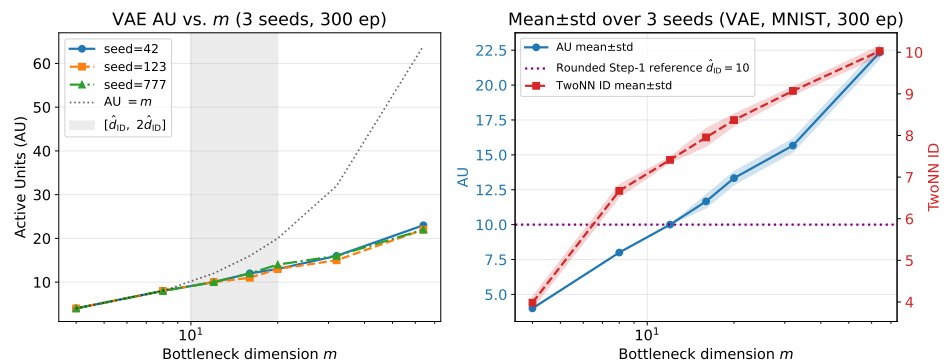


Figure 3. MNIST: multi-seed reproducibility of the AU verification signal (3 seeds, 300 epochs). Left: per-seed $AU(m)$ with the ID-guided search window $[d_{ID}, 2d_{ID}]$ shaded; right: mean $\pm$ std band with the rounded Step-1 TwoNN reference $\hat{d}_{ID} = 10$ (an estimator-derived reference value, not a ground-truth dimension). AU values are highly reproducible ($\sigma \leq 0.5$). Change-point (knee) estimates are deliberately not shown here; they are exploratory only (Appendix B).

5.3.5. Fine-Grained View of the Slowdown Region and a Caution on Change-Point Detection

To inspect the onset of the slowdown more closely, we observed the AU curve on a dense grid with unit steps around $\hat{d}_{ID}$, $m \in \{8, 9, \dots, 16, 18, 20, 24, 28, 32, 40, 48, 64\}$ (17 points, 300 epochs, 3 seeds). $AU(m)$ stays on the diagonal ($AU = m$) up to $m = 8$ – 10 and departs from it at $m \geq 11$; applying a change-point detector based on first-difference ratios (threshold $\tau_{knee} = 0.25$; Appendix B) to this dense grid gives a change point of 10.3 ± 0.9 (per seed: 11, 9, 11), consistent with $\hat{d}_{ID} \approx 10$.

This agreement, however, carries two important qualifications. First, the dense-grid region was placed around the Step-1 TwoNN estimate, so the result is a **local curve summary conditional on a TwoNN-derived search window**, not an independent cross-validation. Second, applying the same detector to the sparse grid $\{4, 8, 12, 16, 20, 32, 64\}$ yields a change point of 43 ± 15 —**automatic change-point detection depends strongly on grid design and should not be operated as a named output of the protocol**. For this reason, Step 3 relies primarily on the robust read-out (multi-seed comparison of the AU values themselves—whether the slowdown region is of the same order as $\hat{d}_{ID}$), and change-point detection is used only as an aid for grid design (detailed sensitivity analysis in Appendix B).

5.3.6. Fashion-MNIST

For Fashion-MNIST, Exp-Real-Geom (Section 5.5; 3 seeds, 300 epochs) gives $\hat{d}_{ID} \approx 10$: TwoNN converges to 9.9–10.1 at $m_{ref} = 5d$ – $10d$, where $d = 9$ is the sweep-grid design value taken from the preliminary 100-epoch estimate (Appendix C). Step 2 then gives the verification point $m_{ver} = 2\hat{d}_{ID} = 20$. Step 3 was run as a 3-seed verification under the same protocol as MNIST (Exp-Fashion-MS; FC-VAE, $\beta = 4$, 300 epochs, $m \in \{4, 8, 12, 16, 20, 32, 64\}$; Table 9).

The result is a **boundary case** that demonstrates the diagnostic capability of the decision rule: all three tested seeds yielded $AU = 7$ at $m = 20$ ($AU(20) = 7.0 \pm 0.0$), so **V1 passes** (activation ratio $0.35 \leq 0.9$; truncation operates robustly) while **the lower V2 bound is violated** ($AU = 7 < \hat{d}_{ID} = 10$; all three tested seeds received the same operational verdict). Following the diagnosis in the third row of Table 2, this suggests either that “ $\beta = 4$ sits on the over-regularizing side for this data–decoder pair” or that “the AE-based Step-1 reference includes variance directions (texture- and style-like factors) that the VAE prunes at this β .” Indeed, the MSE slowdown also occurs at $m \approx 12$ – 16 (Table 9), and the preliminary observation that the AU slowdown region (7–12) is of the same order as $\hat{d}_{ID}$ (Appendix C) is itself replicated over the 3 seeds. Fashion-MNIST is therefore not an

example of the protocol mechanically returning a pass, but a real-data instance in which the pre-registered decision rule detects a boundary case and routes it to the diagnosis of Table 2.

Table 9. Exp-Fashion-MS (Fashion-MNIST FC-VAE, $\beta = 4$, 300 epochs, mean $\pm$ std, 3 seeds). At the pre-specified verification point $m_{\text{ver}} = 20$, V1 passes (activation ratio 0.35) while the lower V2 bound is violated ($\text{AU} = 7.0 < \hat{d}_{\text{ID}} = 10$), with all three tested seeds receiving the same operational verdict—a diagnostic boundary case (third row of Table 2).

m	AU	TwoNN ID (latent)	MSE
4	4.0 ± 0.0	3.71 ± 0.10	0.0184
8	6.3 ± 0.5	5.44 ± 0.19	0.0152
12	7.0 ± 0.0	5.90 ± 0.07	0.0144
16	7.3 ± 0.5	6.10 ± 0.43	0.0141
20	7.0 ± 0.0	5.83 ± 0.17	0.0143
32	9.3 ± 0.5	6.88 ± 0.22	0.0130
64	11.7 ± 0.5	7.79 ± 0.06	0.0120

5.4. Baseline Comparison: Search Cost and Downstream Performance

We quantify the practical value of ID-guided selection against a **full m -grid search** (Exp-Downstream). The evaluation metrics are the test classification accuracy of a linear probe (multiclass logistic regression) on the latent representations and the reconstruction MSE.

AE (reuse of the reference-AE sweep, 300 epochs, seed 42): as shown in the left half of Table 11, the linear-probe accuracy plateaus at $m = 8$ –12 (92.3% at $m = 12$) and does not improve up to $m = 256$ (91.8%; raw 784-dimensional pixels: 91.6%). **The best observed accuracy over the full grid (11 points), 92.3%, lies inside the ID-guided search window [10, 24] (at $m = 12$), and the other in-range points ($m = 16, 20$) are within 1.4 points of it.**

VAE ($\beta = 4$, 300 epochs, 3 seeds): as shown in the right half of Table 11, the same picture holds on the VAE side: the linear-probe accuracy attains its best observed mean value over the grid $\{4, \dots, 64\}$, $89.1 \pm 0.4\%$, at $m = 12$ inside the search window, and is nearly flat for $m \geq 12$. At the VAE-path verification point $m = 20 = 2\hat{d}_{\text{ID}}$ it is $88.2 \pm 0.4\%$ —within 0.9 percentage points of the best observed mean accuracy, although the three-seed sample is too small to support a formal equivalence claim. The AU and MSE values in this table are retrained reproductions of Table 8 (Exp-Mnist-Multi-300) and agree with it—an additional reproducibility check.

Search cost: the full MNIST grid of this paper requires 11 values of $m \times 3$ seeds = 33 runs. The ID-guided path instead performs Step 1 (convergence check over $m_{\text{ref}} \in \{64, 128, 256\}$; 3 single-seed runs) and then either (a) the AE path: training the in-range grid points $\{12, 16, 20\}$ over 3 seeds (9 runs)—12 runs in total ($\approx 64\%$ saved); or (b) the VAE path: training the single point $m = 2\hat{d}_{\text{ID}} = 20$ over 3 seeds and verifying by AU (3 runs)—6 runs in total ($\approx 82\%$ saved). Under the conservative variant that also verifies at the MLE-shifted point $m = 32$ (Section 3.2.3), the VAE path is 9 runs in total ($\approx 73\%$ saved). Note that this accounting counts Step 1 as single-seed runs (this paper quantifies Step-1 uncertainty separately via subsampling CIs and estimator cross-checks); a symmetric accounting that repeats Step 1 over 3 seeds gives 18 runs for the AE path ($\approx 45\%$ saved) and 12 runs for the VAE path ($\approx 64\%$ saved). Table 10 organizes this accounting by condition. The reduction rates compare **numbers of training runs only**; they do not include GPU time, estimator computation, or analysis time. The retained performance is the descriptive comparison given above (the best observed accuracy lies inside the window; the VAE verification point is within 0.9 points of it). This comparison is limited to the MNIST grid configuration and linear probe of this paper and does not establish a general reduction rate (Section 6.1).

Table 10. Training-run accounting for the specific MNIST grid, epoch count, seed design, and verification procedure used in this study. “Runs” counts model trainings only (Step-1 reference-AE trainings included); GPU time, estimator computation, and analysis time are not included.

Path	Full-grid runs	ID-guided runs	Step-1 seeds	Reduction
AE, single-seed Step 1	33	12	1	64%
VAE, single-seed Step 1	33	6	1	82%
VAE + MLE-shifted verification	33	9	1	73%
AE, three-seed Step 1	33	18	3	45%
VAE, three-seed Step 1	33	12	3	64%

Table 11. Exp-Downstream (MNIST): linear-probe test accuracy and MSE vs. m . Left: AE latents (300 epochs, single seed 42; reuse of the reference-AE sweep). Right: FC-VAE ($\beta = 4$) latents (300 epochs, mean $\pm$ std, 3 seeds). Bold m values lie in the ID-guided search window $[10, 24]$, which contains the best observed accuracy over the full grid; for the VAE, $m = 20$ is in addition the pre-specified verification point m_{ver} (Section 3.2.3). Raw-pixel baseline: 91.6%. Because the AE results are based on a single reference-AE seed, the AE probe accuracies should be interpreted descriptively rather than as multi-seed performance estimates.

AE (single seed)			VAE ($\beta = 4$, 3 seeds)			
m	Probe acc. (%)	MSE	m	Probe acc. (%)	AU	MSE
1	60.0	0.0437	4	85.0 ± 0.5	4.0 ± 0.0	0.0282
2	60.0	0.0399	8	87.7 ± 0.1	8.0 ± 0.0	0.0182
4	81.7	0.0292	12	89.1 ± 0.4	10.0 ± 0.0	0.0154
8	91.7	0.0191	16	89.0 ± 0.2	11.7 ± 0.5	0.0138
12	92.3	0.0148	20	88.2 ± 0.4	13.3 ± 0.5	0.0126
16	92.0	0.0121	32	88.1 ± 0.2	15.7 ± 0.5	0.0112
20	90.9	0.0114	64	88.7 ± 0.3	22.3 ± 0.5	0.0089
32	91.2	0.0087				
64	91.8	0.0062				
128	92.2	0.0061				
256	91.8	0.0057				

5.5. Validity Condition of Step 1: What Makes Intrinsic-Dimension Estimation Stable?

Since Step 1 goes through the latents of a reference AE, the focus is whether the geometric distortion of the reference representation corrupts the estimate. In Exp-Real-Geom we swept four models with different isometry properties (plain AE / IsometricAE [21] / CAE [11] / DAE [10]) on MNIST and Fashion-MNIST over $m_{\text{ref}} \in \{d, 2d, 3d, 5d, 10d\}$ (d is the grid design value from preliminary estimates: 10 for MNIST, 9 for Fashion-MNIST), comparing TwoNN, MLE, κ , Trustworthiness, and kNN overlap (3 seeds, 300 epochs).

Table 12. Exp-Real-Geom (MNIST, mean $\pm$ std, 3 seeds): TwoNN through four reference models. Convergence is driven by the bottleneck margin $m_{\text{ref}}/\hat{d}_{\text{ID}}$, largely independently of the isometry (condition number κ).

Model	TwoNN ($m_{\text{ref}} = d$)	TwoNN ($m_{\text{ref}} = 5d$)	TwoNN ($m_{\text{ref}} = 10d$)	κ ($m_{\text{ref}} = 10d$)
AE	7.3 ± 0.4	10.0 ± 0.1	10.0 ± 0.1	4.1×10^7
IsometricAE	7.5 ± 0.4	10.1 ± 0.1	10.1 ± 0.1	7.8
CAE	7.2 ± 0.5	9.9 ± 0.1	10.0 ± 0.1	3.2×10^3
DAE	7.6 ± 0.4	10.0 ± 0.1	10.0 ± 0.1	6.3×10^3

The results (Table 12) are clear: (1) in all four models, TwoNN converges from the underestimate at $m_{\text{ref}} = d$ (≈ 7.3 – 7.6) to ≈ 10 at $m_{\text{ref}} \geq 5d$, and the inter-model differences (within ± 0.3) are an order of magnitude smaller than the effect of increasing m_{ref} (about 2.5 points); (2) IsometricAE improves κ from 10^7 to around 8, yet its effect on TwoNN stays within ± 0.2 . The same pattern was obtained on Fashion-MNIST. That is, **under the datasets**,

architectures, sample sizes, and estimators tested here, the dominant condition for Step-1 stability is not “improved isometry” but the “bottleneck margin $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ ” (consistent with the δ_T -dominance implication of the conditional-stability rationale in Section 4.3), justifying the design of Step 1 with a plain reference AE at $m_{\text{ref}} = 256 \gg \hat{d}_{\text{ID}}$ (Figure 4).

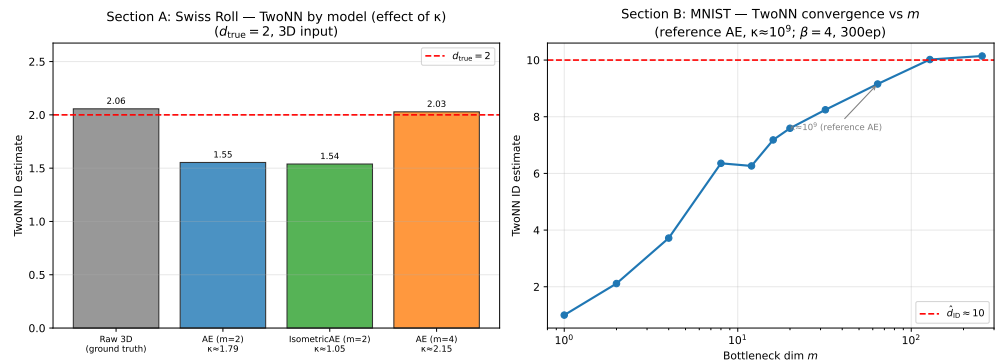


Figure 4. Step-1 stability: (left) on the Swiss Roll, TwoNN recovers once $m_{\text{ref}} > d_{\text{true}}$ across the tested reference models, despite large differences in the decoder-Jacobian condition number; (right) on MNIST, TwoNN stabilizes for $m_{\text{ref}} \geq 48 \approx 5\hat{d}_{\text{ID}}$ even where the condition number κ grows sharply. This result does not establish that isometry is generally irrelevant to intrinsic-dimension estimation.

5.6. Applicability Boundary: Conv-VAEs and Natural Images

This section is not a “report of failures” of the protocol but a quantification of the conditions under which the failure modes of Table 2 actually occur.

5.6.1. Conv-VAE: Truncation Failure Due to Insufficient β (3 Seeds)

We swept an MNIST Conv-VAE (2 Conv layers + 2 ConvTranspose layers, KL cyclic annealing with $\beta_{\text{max}} = 4$ [27], 150 epochs, 3 seeds) over $m \in \{4, 8, 16, 32, 64, 96, 128, 192, 256\}$ (Exp-Conv-M256). For $m \leq 64$, $\text{AU} = m$ (no truncation) persists in all seeds; AU falls below m for the first time at $m = 96$ ($\text{AU} = 90.7 \pm 1.2$), yet the activation rate remains as high as 94% at $m = 256$ (Figure 5). Extending to $m \in \{384, 512\}$ (Exp-Conv-M512) still keeps $\text{AU} \approx m$ —that is, at $\beta_{\text{max}} = 4$ the experiments do not observe substantial truncation within $m \leq 512$; under the surrogate interpretation, this pattern is consistent with an effective truncation threshold beyond the tested range ($d_{\text{eff}}^*(\beta) > 512$)—and the clear $\text{AU} = m$ pattern flags the diagnostic condition “no substantial truncation observed within the tested range,” which is consistent with β being too small relative to the decoder capacity (first row of Table 2).

Two caveats frame the surrogate-based interpretation below. First, the surrogate analysis of Proposition 2 assumes a fixed β , whereas the convolutional experiments use cyclical KL annealing (three cycles; within each cycle, β increases linearly from 0 to β_{max} over the first half and is held at β_{max} over the second half [27]); β_{max} is therefore a training-schedule parameter and should not be identified with the fixed β appearing in Proposition 2. Second, the discussion of τ_{eff}^2 below is a post-hoc heuristic interpretation intended to connect the observed behavior with the linear surrogate; it is not an independently estimated physical or probabilistic parameter of the nonlinear decoder. From the viewpoint of the surrogate condition (Proposition 2), a powerful convolutional decoder makes the effective noise variance τ_{eff}^2 small (a back-computed calibration ≈ 0.30 for the FC-VAE and $\ll 0.30$ for the Conv-VAE; a heuristic extrapolation), lowering the threshold $\beta\tau_{\text{eff}}^2$ and inflating $d^*(\beta)$. This interpretation is consistent with the β -strengthening experiments: at $\beta_{\text{max}} = 10$, $\text{AU} = 95.7 \pm 5.2$ (activation ratio 37%) at $m = 256$; at $\beta_{\text{max}} = 20$, $\text{AU} = 68.0 \pm 2.2$ (27%)—a clear reduction in the activation ratio and evidence of stronger truncation within the tested range, although stable AU saturation (a plateau in m) was not established (Exp-ConvV-

HiBeta; **3 seeds**, details in Appendix C). This is consistent with the qualitative prediction that increasing β moves the truncation to lower dimensions, identifying “strengthening β according to data and architecture” as a necessary condition for applying the protocol to Conv-VAEs.

Verdict (MNIST Conv-VAE): Step 1—usable (reusing $\hat{d}_{ID} \approx 10$ from the FC reference AE); Step 2—search window obtained ($m \geq 20$); Step 3—**fail** at $\beta_{\max} = 4$ (activation ratio $0.94 > 0.9$ at $m = 256$; V1 violated). At $\beta_{\max} \in \{10, 20\}$, V1 holds (activation ratios 0.37 and 0.27; all three tested seeds received the same operational verdict) but V2 remains violated since $AU = 95.7 \pm 5.2$ and $68.0 \pm 2.2 > 3\hat{d}_{ID} = 30$; we judge this as “truncation induced, but order consistency with $\hat{d}_{ID}$ not yet achieved.”

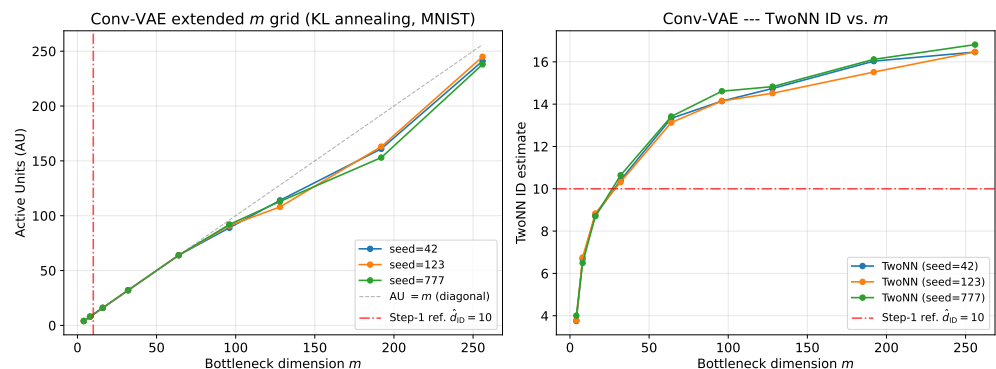


Figure 5. Boundary condition (Exp-Conv-M256): the MNIST Conv-VAE with $\beta_{\max} = 4$ keeps $AU = m$ (no truncation) up to $m = 64$ and stays at 94% activation even at $m = 256$ —the failure mode “ β too small relative to decoder power” detected by the AU signal. The reference value $\hat{d}_{ID} = 10$ was obtained from the independently trained reference-AE procedure of Step 1, not from the Conv-VAE latents shown here.

5.6.2. Natural Images: Bounded Trust Region of Step 1 and Unmet Step 3 (3 Seeds)

For Conv-VAEs on CIFAR-10 [28] and SVHN [29] (3 Conv layers, $\beta_{\max} = 4$, 150 epochs, 3 seeds, $m \in \{16, 32, 64, 128, 256\}$), two observations follow (Table 13). First, **Step 1 works conditionally**: the latent TwoNN of CIFAR-10 at $m = 64 \approx 2\hat{d}_{ID}$ is 25.21 ± 0.41 , consistent with the raw-input estimate (25.20) and with Pope et al. [6] (≈ 26). Beyond $m \geq 128$, however, it inflates monotonically to $28.8\text{--}30.9$ —**on natural images the estimate is consistent with prior estimates only near $m_{\text{ref}} \approx 2\hat{d}_{ID}$, and “the larger the better” does not hold** (the true ID is unknown, so nothing stronger than consistency can be claimed; fourth failure mode of Table 2). For the natural-image experiments considered here, $m_{\text{ref}} \approx 2\hat{d}_{ID}$ thus behaved as a **provisional consistency region**, not as a validated selection rule. The contrast with MNIST (stable even at $m_{\text{ref}} = 25\hat{d}_{ID}$) is attributed to latent-geometry distortion while all dimensions remain in use ($AU = m$, no truncation). For SVHN the discrepancy is even larger (latent 27.1 vs. raw 10.1 at $m = 256$). This discrepancy should not by itself be interpreted as latent-space inflation: the raw-input estimate is not a ground truth either, as it may be affected by ambient-dimensional distance concentration and by heterogeneous local geometry (Section 2), so the two values are best read as two differently biased views of an unknown quantity.

Why is the distortion so pronounced on natural images (hypotheses)? We regard the gap between MNIST and natural images not as a matter of degree but as reflecting a qualitative difference in data geometry. **The following mechanisms are hypotheses consistent with the observed pattern, not experimentally distinguished explanations.** First, **manifold entanglement**—operationally, a distribution structured as a union of sub-manifolds whose local intrinsic dimension varies strongly across classes and regions, in conflict with the single-global-metric assumption of TwoNN: MNIST consists of centered,

size-normalized single objects whose dominant factor is smooth stroke deformation, and is therefore well approximated by a single smooth manifold; natural images are compositions of factors at different scales—object category, pose, texture, lighting, background—and behave like such a union of submanifolds (Pope et al. [6] likewise report large inter-class variation of the ID, and Allegra et al. [16] provide a related treatment of heterogeneous local intrinsic dimensionality). For TwoNN, which assumes a single global metric, such inhomogeneous structure tends to degrade the preservation of neighbor ranks (the δ_T of Section 4.3) as m_{ref} grows. Second, **high-frequency texture and the effective noise variance**: reconstructing natural images requires encoding fine texture, and a powerful Conv decoder drives τ_{eff}^2 extremely small. Under the surrogate interpretation, this pattern would correspond to an effective active-direction threshold larger than the tested m -range ($d^*(\beta) \gg m$), resulting in little or no observable truncation; in a latent space where all coordinates remain in use, the dominant manifold factors and the texture residuals are mixed within one metric and the neighborhood structure is distorted—consistent with the pattern observed in our experiments, where $AU = m$ and TwoNN inflation occur simultaneously. Third, **the need for hierarchical latent structure**: generative factors of natural images form a scale hierarchy from global (objects) to local (texture), and the architectural mismatch of forcing this hierarchy into a single flat latent vector with an isotropic prior may itself be a source of distortion. Under this hypothesis, a natural next step is to apply the protocol per level to the scale-wise latents of hierarchical VAEs [30] (reading the AU of each level separately) (Section 6.4). All three hypotheses are consistent with our observations (persistent $AU = m$, m_{ref} -dependent TwoNN inflation, partial recovery under β -strengthening), but a direct disambiguation is left for future work. Second, **Step 3 is unmet**: no AU saturation appears for $m \leq 256$ (only $AU < m$ from $m \geq 128$ on SVHN); AU verification requires strengthened β , restricted decoder capacity, and exploration of larger m ranges (a preliminary observation of plateau-like behavior at $\beta_{\text{max}} = 80$ with a thin decoder is reported in Appendix C).

Verdict (natural images): Step 1—usable only in the provisional consistency region $m_{\text{ref}} \approx 2\hat{d}_{\text{ID}}$ (CIFAR-10: 25.21 ± 0.41 , consistent with reported values; inflation beyond); Step 2—search window obtained conditionally (CIFAR-10: $m \geq 50$); Step 3—**unresolved** (CIFAR-10: activation ratio 1.0 at $m = 256$, V1 violated; SVHN: activation ratio 0.81 satisfies V1 but $AU = 206.7 \gg 3\hat{d}_{\text{ID}}$ violates V2).

Table 13. Natural images (Exp-Nat-Cifar / Exp-Nat-SVHN): Conv-VAE with KL cyclic annealing ($\beta_{\text{max}} = 4$, 150 epochs, mean $\pm$ std, 3 seeds). For CIFAR-10, Step 1 agrees with Pope et al. [6] only near $m \approx 2\hat{d}_{\text{ID}}$ (for SVHN the latent–raw discrepancy is larger; see text); Step 3 (AU saturation) is not reached in the tested range.

m	CIFAR-10		SVHN	
	AU	TwoNN ID	AU	TwoNN ID
16	16.0 ± 0.0	13.12 ± 0.12	16.0 ± 0.0	11.30 ± 0.12
32	32.0 ± 0.0	19.26 ± 0.23	32.0 ± 0.0	15.08 ± 0.08
64	64.0 ± 0.0	25.21 ± 0.41	64.0 ± 0.0	19.97 ± 0.12
128	128.0 ± 0.0	28.83 ± 0.41	125.3 ± 0.5	24.60 ± 0.36
256	256.0 ± 0.0	30.91 ± 0.64	206.7 ± 1.2	27.08 ± 0.34
Raw-input TwoNN = 25.20, MLE = 26.56			Raw-input TwoNN = 10.09, MLE = 18.85	

6. Discussion

6.1. What Differs from Conventional Model-Selection Criteria?

The conventional means of choosing m in practice are m -sweeps of validation reconstruction error or ELBO, or downstream task performance. These tell us “the m at which performance plateaus,” but (i) they require sweeping the entire grid, and (ii) they do not

distinguish the cause of the plateau (data manifold dimension, model capacity, or optimization). Our protocol does not replace them but **complements** them: Step 1 narrows the search window to the vicinity of $\hat{d}_{\text{ID}}$ before any sweep (reducing search cost), and the AU of Step 3 reports—independently of performance—how many dimensions are actually in use, directly visualizing the presence of redundant dimensions. Indeed, on MNIST the MSE verification, the AU verification, $\hat{d}_{\text{ID}}$, and the linear-probe accuracy all supported the same search window [10, 24] (Section 5.4), and on the MNIST grid used in this paper, choosing m inside the window reduced the number of training runs by roughly 60–80% relative to the full grid search while nearly retaining the downstream classification accuracy (no general reduction rate is claimed). This comparison is, however, limited to a linear probe on MNIST; systematic comparison with nonlinear downstream tasks and disentanglement metrics remains future work.

6.2. Stability vs. Validity: What Does $\hat{d}_{\text{ID}}$ Estimate?

A distinction maintained throughout this paper is that “TwoNN/MLE being **numerically stable** (low variance across seeds and m_{ref})” is different from “being **unbiased** with respect to the true intrinsic dimension.” The latent space of the reference AE is far from isometric (κ reaches 10^7 ; Table 12), and what TwoNN estimates is, strictly speaking, “the local degrees of freedom of the reference representation.” The MNIST value $\hat{d}_{\text{ID}} \approx 10$ –12 is supported by several pieces of indirect evidence—(i) convergence over multiple m_{ref} , (ii) TwoNN–MLE agreement, (iii) agreement across four reference models (Table 12), (iv) narrow subsampling CIs and order-of-magnitude agreement with estimators of different principles (DANCo, ESS; Table 7), and (v) consistency with reported values on CIFAR-10—but this is not a direct proof of unbiasedness. The protocol therefore operates $\hat{d}_{\text{ID}}$ as “a reference value giving the center of the search window” and closes the loop with the Step-3 verification. A remaining task is the calibration of $\hat{d}_{\text{ID}}$ against ground truth beyond synthetic data (a systematic evaluation of estimator bias).

6.3. Limits of AU Verification: Why Change-Point Detection Is Not a Named Output

An earlier version of this framework (the predecessor of this work) treated the AU slowdown point m^{knee} as one of its diagnostics; this paper removes it from the named outputs. The reasons, shown in Section 5.3.5, are that (i) the sparse-grid detection (43 ± 15) is dominated by the grid interval, (ii) stabilization requires a TwoNN-derived search window and hence is not independent validation, and (iii) the detection is also sensitive to the choice of detector (threshold vs. second difference) (Appendix B). The robust part of AU verification is the read-out “do the AU values themselves (multi-seed, $\sigma \leq 0.5$) slow down or saturate at the same order as $\hat{d}_{\text{ID}}$?”—the protocol is designed to use only this robust part in its decision rule (Equations (2)–(3), which involve no change-point detection). Nested validation, in which the search-window estimation and the change-point evaluation are performed on disjoint seeds or data subsets, remains future work.

6.4. Limitations and Future Work

- **Validation scope:** multi-seed validation of the full protocol is limited to synthetic manifolds and the MNIST and Fashion-MNIST AU verifications (the MNIST Step-1 sweep is single-seed with separately quantified uncertainty—a full multi-seed reference-AE sweep over all m_{ref} values was not performed; the Fashion-MNIST Step 3 is a diagnostic detection example with V1 passing and the lower V2 bound violated; Section 5.3.6). In addition, all experiments use fixed-order data subsets (Section 5.1), so the results are not averages over independent data partitions. The recovery of Conv-VAE truncation by strengthened β (Exp-ConvV-HiBeta) was replicated over 3 seeds (V1 holds, V2

unmet; Section 5.6), but achieving order consistency requires joint control of β and decoder capacity.

- **Step 3 on natural images:** stable identification of AU saturation is unmet; dense sweeps of $\beta_{\max} \in [40, 120]$, decoder-capacity control, and improved seed stability are future work (Appendix C). The hypotheses stated in Section 5.6 (manifold entanglement, texture-driven shrinkage of τ_{eff}^2 , and the need for hierarchical latent structure) also suggest directions for improvement: (i) applying the protocol per level to the scale-wise latents of hierarchical VAEs [30], reading the AU truncation at the level of global factors; (ii) class-conditional or region-wise estimation of $\hat{d}_{\text{ID}}$ per submanifold (mitigating the inhomogeneity of the union structure); (iii) bounding τ_{eff}^2 from below via decoder capacity or rate-allocation control (separating texture residuals from the AU diagnosis). These extensions keep the estimate-set-verify framework of the protocol itself and refine Steps 1 and 3 to operate per level or per subset.
- **β dependence and decision thresholds:** the AU read-out depends on β (AU patterns confirmed stable for $\beta \in [2, 8]$; Appendix D), and AU is coordinate-dependent (Section 3.1), so its read-out is tied to the chosen prior and latent parameterization. Actively exploiting the β dependence of $d^*(\beta)$ —a “spectral read-out by β sweep”—is an interesting extension. The thresholds of the decision rule (activation ratio 0.9, factor 3) are working defaults that separate the passing and failing cases of this paper by a wide margin; calibrating them on more diverse data is future work.
- **Scope of the theory:** Proposition 2 is a linear-Gaussian approximation, and its non-linear extrapolation rests on post-hoc consistency checks via τ_{eff}^2 . Deriving activation conditions in nonlinear settings is open.
- **Computational cost:** the multi-seed m -sweep of Step 3 takes hours to a day of GPU time. Step 1 shrinks the sweep range, but further lightening (e.g., early-epoch AU prediction) is future work.

7. Conclusions

This study introduced a **conditional estimate-set-verify protocol** for selecting AE and VAE bottleneck dimensions: (S1) intrinsic-dimension estimation through reference representations with $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$, (S2) setting m as $\hat{d}_{\text{ID}}$ plus a topological margin (AE: $m \in [\hat{d}_{\text{ID}}, 2\hat{d}_{\text{ID}}]$; VAE: a generous candidate $m \geq 2\hat{d}_{\text{ID}}$), and (S3) post-training verification by AU and MSE equipped with the screening rule V1/V2. The protocol uses intrinsic-dimension estimates to define a candidate range and uses AU and reconstruction-error behavior to diagnose whether the selected VAE bottleneck was effectively truncated. The synthetic experiments demonstrated that the procedure can detect both ordinary saturation and a topological boundary case in which the default margin is insufficient ($\text{AU}_{\text{sat}} = d_{\text{true}}$ or d_{emb} under sufficient margin— $m \geq 5$ on the Torus—with the boundary over-pruning at $m \leq 4$ caught by the MSE cross-check; 3–5 seeds, $\sigma = 0$). On MNIST, the protocol produced a consistent multi-seed diagnostic under the tested fully connected architecture (the primary Step-1 reference $\hat{d}_{\text{ID}} = 10$, with the candidate-VAE latent cross-check 10.03 ± 0.12 consistent with the AU slowdown region; 3 seeds) and reduced the number of training runs by roughly 60–80% relative to the specific full-grid experiment while containing the best observed linear-probe accuracy inside the window. On Fashion-MNIST, it correctly identified a lower-bound violation of the AU consistency criterion rather than returning an unconditional pass.

These results should not be interpreted as establishing that $\hat{d}_{\text{ID}}$ estimates the topological embedding dimension, or that AU estimates the intrinsic dimension. The former is estimator- and representation-dependent, while the latter depends on the regularization strength, the AU threshold, the decoder capacity, and the architecture; in particular, the

linear-Gaussian activation condition $\lambda_j > \beta\tau^2$ is a theoretical surrogate, not a quantitative theorem for nonlinear VAEs. The quantified applicability boundaries make the conditional nature concrete: under the datasets, architectures, and estimators tested, intrinsic dimension estimation was numerically stable only with bottleneck margin $m_{\text{ref}} \gg \hat{d}_{\text{ID}}$ (isometry improvements were secondary; four-model comparison), inflates beyond $m_{\text{ref}} \approx 2\hat{d}_{\text{ID}}$ on natural images, and for the tested Conv-VAE schedule with $\beta_{\text{max}} = 4$ sufficient truncation cannot be assumed (a V1 violation with $\text{AU} = m$); strengthening β_{max} induces truncation but order consistency with $\hat{d}_{\text{ID}}$ (V2) remains unmet (3 seeds). Future work should therefore focus on calibrated decision thresholds, independent data and seed splits (including multi-seed reference-model sweeps), multi-seed stable identification of Step 3 on natural images and Conv-VAEs, nested validation of change-point detection, bias calibration of $\hat{d}_{\text{ID}}$, and theory for nonlinear posterior collapse.

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Data Availability Statement: MNIST [23], Fashion-MNIST [24], CIFAR-10 [28], and SVHN [29] are publicly available. Anonymous source code, configuration files, data-subset indices, random seeds, seed-specific training logs, figure-generation scripts, and the numerical results supporting all tables are provided as supplementary material for peer review. Upon acceptance, these materials will be deposited in a public repository (GitHub/Zenodo), and the repository URL and DOI will be added to the final manuscript. The repository will include the scripts required to reproduce the AU, TwoNN/MLE, reconstruction-error, and grid-sensitivity analyses reported in this paper.

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Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
VAE	Variational Autoencoder
AU	Active Units
ID	Intrinsic Dimension
TwoNN	Two-Nearest-Neighbor (ID estimator)
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
PCA	Principal Component Analysis
KL	Kullback–Leibler (divergence)
FC	Fully Connected
Conv	Convolutional
CAE	Contractive Autoencoder
DAE	Denoising Autoencoder
CKA	Centered Kernel Alignment

Appendix A. Proofs and Derivations

Appendix A.1. Derivation Sketch of Proposition 2 (Surrogate Activation Condition)

Consider a β -VAE with linear encoder $f(x) = W_e x$, linear decoder $g(z) = W_d z$, prior $p(z) = \mathcal{N}(0, I_m)$, and likelihood $p(x|z) = \mathcal{N}(W_d z, \tau^2 I_N)$. Taking the input in its PCA basis with covariance $\Sigma = \text{diag}(\lambda_1, \dots, \lambda_N)$ ($\lambda_1 \geq \lambda_2 \geq \dots$), the β -ELBO separates across eigendirections in the linear-Gaussian model. Let $R_j \geq 0$ be the rate (KL contribution) allocated to eigendirection j , and let $D_j(R_j)$ denote its **residual variance contribution** (the distortion remaining on that direction); under the Gaussian rate–distortion relation, $D_j(R_j) = \lambda_j e^{-2R_j}$. Here R_j is measured in nats. The relevant term of the β -ELBO is

$$\mathcal{L}_j(R_j) = -\frac{1}{2\tau^2} \lambda_j e^{-2R_j} - \beta R_j + \text{const.}, \quad (\text{A1})$$

written up to additive constants and **maximized** with respect to $R_j \geq 0$ (λ_j is the j -th eigenvalue of the centered input covariance and τ^2 the observation-noise variance of the Gaussian likelihood). Maximization over R_j is a concave program, and the KKT condition $\partial \mathcal{L}_j / \partial R_j = 0$ yields $R_j^* = \max(0, \frac{1}{2} \log(\lambda_j / \beta \tau^2))$. The activation condition (Equation (4)) is $R_j^* > 0 \iff \lambda_j > \beta \tau^2$. This coincides with Shannon’s water-filling theorem [31] with water level $\beta \tau^2$, and reduces to the known PPCA result [17] for $\beta = 1$. With m available coordinates the optimum activates the top $\min(m, d^*(\beta))$ directions, giving claims (2) and (3).

Content of the approximation: the derivation replaces the expected KL term $\mathbb{E}_x[\text{KL}(q_\phi(z|x) \| p(z))]$ by the mutual information $I(x; z)$ (the rate). Their difference is given exactly by the so-called ELBO-surgery decomposition [32]

$$\mathbb{E}_x[\text{KL}(q_\phi(z|x) \| p(z))] = I(x; z) + \text{KL}(q_\phi(z) \| p(z)), \quad (\text{A2})$$

i.e., the aggregated-posterior mismatch $\text{KL}(q_\phi(z) \| p(z)) \geq 0$, and the proposition holds only where this term is negligible.

Behavior of the mismatch term (implications for nonlinear optimization): we briefly organize how the dropped mismatch term behaves. (i) For the particular globally optimized linear-Gaussian solution considered here, the approximation is **exact on truncated coordinates**: there $q_\phi(z_j|x) = p(z_j)$, so those coordinates contribute nothing to $\text{KL}(q_\phi(z) \| p(z))$. On **active coordinates**, the aggregated-posterior mismatch is **assumed to be negligible** at this optimum: the marginal variance of the posterior mean plus the posterior variance equals the prior variance (a property of PPCA-type optima [3,17]), and this variance-matching property motivates the assumption but does not by itself establish exact vanishing. This assumption is not made for arbitrary linear or nonlinear VAEs. (ii) During the training of a nonlinear VAE the term is nonnegative in general; it is small early in training (posteriors close to the prior) and can grow as coordinates activate and the aggregated posterior develops cluster structure (multimodality). (iii) By Equation (A2), the larger the mismatch term, the larger the actual KL penalty paid for the same mutual information (rate)—that is, in nonlinear settings β is effectively amplified, and the threshold prediction $\lambda_j > \beta \tau^2$ tends to **predict weaker truncation** than would be induced by the additional aggregated-posterior mismatch; equivalently, it may overestimate the number of active dimensions relative to the nonlinear model. This direction is consistent with how the surrogate is used in this paper: the failure of truncation observed for Conv-VAEs and natural images ($\text{AU} = m$; Section 5.6) cannot be explained by the mismatch term (which only works toward stronger truncation) and requires the separate mechanism of τ_{eff}^2 shrinkage by decoder capacity—this is the basis of the interpretation in Section 5.6. Conversely, the observation that on MNIST the AU slows down gently at a level somewhat

above $\hat{d}_{\text{ID}}$ (Section 5.3) is qualitatively compatible with the mismatch term growing with cluster structure and pushing truncation to the conservative side. A direct check of this reading, by systematically measuring $\text{KL}(q_\phi(z) \parallel p(z))$ during training, is future work. The gradient dynamics of trained nonlinear VAEs are not treated, so the implications for nonlinear settings are qualitative.

Appendix A.2. Supporting Argument for the Conditional-Stability Rationale of Section 4.3 (TwoNN)

By the bi-Lipschitz condition, each distance changes by a factor within $(1 \mp \varepsilon)^2$. The TwoNN statistic r_2/r_1 is a ratio of distances, so uniform scaling cancels and only the non-isometric component ($\varepsilon \neq 0$) distorts the ratio. Trustworthiness $\geq 1 - \delta_T$ guarantees rank preservation of the two nearest neighbors with high probability, probabilistically suppressing large variations of r_2/r_1 due to rank reversals. Under both conditions, the change (EMD) of the empirical distribution of r_2/r_1 is bounded by a continuous function $C(\varepsilon, \delta_T)$, and so is the deviation of the TwoNN estimate ($C = 0$ at $\varepsilon = \delta_T = 0$). This is a finite-sample, finite-dimension approximation; rigorous constants depend on the specific setting.

Appendix A.3. Other Interpretive Propositions

Pullback metric and local isometry: the decoder g is locally isometric if and only if $G(z) = J_g(z)^\top J_g(z) = I_m$. The condition number $\kappa = \sigma_{\max}(J_g)/\sigma_{\min}(J_g)$ measures anisotropy; $\kappa = 1$ implies only isotropy, not isometry. Moreover, κ is a local metric diagnostic and does not fully characterize the neighbor-rank distortions that drive TwoNN. In this paper κ is used as an auxiliary indicator of distortion.

Effective rank of the decoder Jacobian: when an AE of sufficient capacity reconstructs a manifold $\mathcal{M}$ exactly, the number of significant singular values of $J_g(z)$ equals $\min(m, d_{\text{emb}})$ (an idealization neglecting finite capacity and optimization error).

Appendix B. Exploratory Analysis of AU Change-Point Detection

This appendix collects the sensitivity data behind the decision (Section 6.3) **not** to make the automatic detection of a change point (knee) m^{knee} on the AU curve a named protocol output.

Appendix B.1. Definition

On a grid $\mathcal{M} = \{m_1 < \dots < m_K\}$, define the AU increase rate $s(m_i) = [\text{AU}(m_i) - \text{AU}(m_{i-1})]/(m_i - m_{i-1})$ and the normalized rate $\rho(m_i) = s(m_i)/\max_j s(m_j)$, and set

$$m^{\text{knee}} = \min\{m_i \in \mathcal{M} : \rho(m_i) < \tau_{\text{knee}}\}, \quad \tau_{\text{knee}} = 0.25 \quad (\text{A3})$$

(first-difference ratio method). An alternative is the second-difference (d2) method, which takes the point of maximum absolute discrete curvature of $\text{AU}(m)$.

Appendix B.2. Grid Sensitivity

In an idealized monotone-saturation model (AU saturates stepwise at $d^*(\beta)$ without noise), $|m^{\text{knee}}(\mathcal{M}) - d^*(\beta)| \leq \max_i (m_{i+1} - m_i)$ holds (the detection error is bounded by the maximum grid interval). On the actual MNIST FC-VAE ($\beta = 4$, 300 epochs, 3 seeds), stochastic fluctuations of the empirical AU curve produce variation beyond this bound (Table A1).

Table A1. Grid sensitivity of the change-point detector on the MNIST FC-VAE ($\tau_{\text{knee}} = 0.25$, 3 seeds). The sparse-grid value is dominated by the grid interval; the dense grid is placed around the TwoNN estimate and is therefore a TwoNN-conditioned curve summary, not independent validation.

Condition	Max. interval	seed 42	seed 123	seed 777	Mean $\pm$ std
Sparse grid, 150 ep	32	64	32	64	53 ± 15
Sparse grid, 300 ep	32	64	32	32	43 ± 15
Dense grid, 300 ep	16 (step 1 in 8–16)	11	9	11	10.3 ± 0.9

Extending the epochs ($150 \rightarrow 300$) has little effect; the dominant factors are grid design and seed, not insufficient convergence. A single-seed preliminary observation on a different grid also gave $m^{\text{knee}} = 12$ (Exp-Mnist-Base), making the grid/initialization dependence explicit.

Appendix B.3. Detector and Threshold Sensitivity

Table A2 compares the τ -method and the d2 method on the dense-grid data. The d2 method is sensitive to AU fluctuations in the transition region (e.g., $8 \rightarrow 10 \rightarrow 9 \rightarrow 11 \rightarrow 10 \rightarrow 11 \rightarrow 13$) and deviates strongly for seed 123 ($m^{\text{knee}} = 28$). The value $\tau_{\text{knee}} = 0.25$ is a working default of this study; on Fashion-MNIST a boundary sensitivity (the detected value changes around τ) has been confirmed—we claim no general meaning for τ_{knee} .

Table A2. Detector sensitivity on the dense grid (300 ep, 3 seeds).

Method	seed 42	seed 123	seed 777	Mean $\pm$ std
τ -method ($\tau = 0.25$)	11	9	11	10.3 ± 0.9
Second difference (d2)	15	28	12	18.3 ± 6.8
Reference: $\hat{d}_{\text{ID}} \approx 10$ (TwoNN, 3-seed mean)				

Appendix B.4. Conclusion

Change-point detection is sensitive to grid, seed, detector, and threshold, and its stabilization requires the circular condition of a TwoNN-derived search window. The protocol therefore confines change points to an aid for grid design (where to sample densely around $\hat{d}_{\text{ID}}$), and bases its verification verdicts on multi-seed comparison of the AU values themselves. Nested validation separating window estimation from change-point evaluation remains future work.

Appendix C. Details of Exploratory and Boundary Experiments

Within this appendix, Exp-ConvV-HiBeta is a **3-seed (42, 123, 777) boundary experiment** that supports the verdicts of Section 5.6; the remaining experiments are **single-seed (42) or partial exploratory observations** and are not used as direct evidence for the main-text claims.

Appendix C.1. High- β Induction for the Conv-VAE (Exp-ConvV-HiBeta)

Table A3 shows the effect of strengthening β_{max} for the MNIST Conv-VAE (KL cyclic annealing, 150 epochs, $m \in \{16, 32, 64, 96, 128, 192, 256\}$, 3 seeds; Exp-ConvV-HiBeta-MS). At $\beta_{\text{max}} = 10$ the activation rate drops to 37% at $m = 256$; at $\beta_{\text{max}} = 20$ the truncation moves further down. The behavior is consistent across the 3 seeds (the inter-seed standard deviation of AU at $m = 256$ is about 5), in line with the qualitative prediction of the surrogate condition that increasing β moves the truncation to lower dimensions (Figure A1).

Table A3. Exp-ConvV-HiBeta (mean $\pm$ std, 3 seeds): AU of the MNIST Conv-VAE under strengthened $\beta_{\max}$ (mean activation rate in parentheses). The $\beta_{\max} = 4$ column is the 3-seed Exp-Conv-M256 reference.

m	$\beta_{\max} = 4$	$\beta_{\max} = 10$	$\beta_{\max} = 20$
32	32.0 $\pm$ 0.0 (100%)	32.0 $\pm$ 0.0 (100%)	26.7 $\pm$ 0.5 (83%)
64	64.0 $\pm$ 0.0 (100%)	52.3 $\pm$ 3.1 (82%)	39.3 $\pm$ 2.5 (61%)
128	111.7 $\pm$ 2.6 (87%)	72.3 $\pm$ 2.5 (57%)	52.7 $\pm$ 0.5 (41%)
256	241.3 $\pm$ 2.9 (94%)	95.7 $\pm$ 5.2 (37%)	68.0 $\pm$ 2.2 (27%)

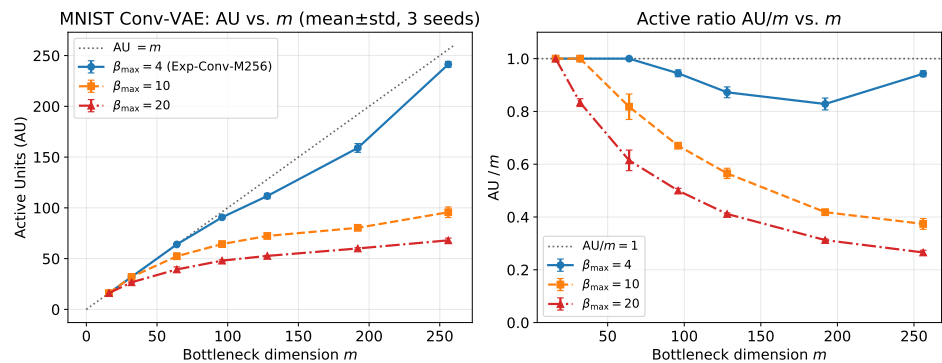


Figure A1. Exp-ConvV-HiBeta (mean $\pm$ std, 3 seeds): increasing $\beta_{\max}$ shifts the AU truncation to lower dimensions, consistent with the surrogate activation condition.

Appendix C.2. Search for AU Plateaus on Natural Images (Exp-NatImg-Sweep)

In a sweep of the CIFAR-10 Conv-VAE combining $m \in [32, 1024]$, $\beta_{\max} \in \{10, 20, 40, 80\}$, and decoder capacities (Standard/Deep/Thin), plateau-like behavior of AU around ≈ 24 was observed over $m \in [512, 1024]$ under the condition $\beta_{\max} = 80$ with a thin decoder (preliminary evidence). Across all conditions the latent TwoNN remained stable at 24.7–25.4, consistent with Pope et al. [6]. Stable identification including seed and grid stability is unmet and remains future work.

Appendix C.3. Preliminary VAE Observation on Fashion-MNIST (Exp-Fashion)

This is the preliminary observation that preceded the 3-seed verification (Exp-Fashion-MS) of Section 5.3.6. For the Fashion-MNIST VAE ($\beta = 4$, 100 epochs, single seed), the AE side TwoNN converges to ≈ 9.4 for $m_{\text{ref}} \geq 32$ (the grid-design value $d = 9$ of Exp-Real-Geom is based on this preliminary estimate), and an AU slowdown region of the same order as $\hat{d}_{\text{ID}} \approx 9$ –10 was observed (same tendency as MNIST). This preliminary observation has been superseded by Exp-Fashion-MS (3 seeds, 300 epochs; Section 5.3.6), where a V1 pass with a violation of the lower V2 bound was confirmed consistently across the 3 seeds.

Appendix C.4. AU_{sat} on Topological Manifolds (Exp-Syn-Top)

On the Torus and the Möbius band (both $d_{\text{ID}} = 2$, $d_{\text{emb}} = 3$) we observed $AU_{\text{sat}} = 3 = d_{\text{emb}}$, and on the circle S^1 ($d_{\text{emb}} = 2$) $AU_{\text{sat}} = 2$ (for $m \geq 6$, conditional). The AU_{sat} of the Swiss Roll depends on preprocessing (2 with standardization, 3 otherwise), so the main evidence for the $AU_{\text{sat}} = d_{\text{emb}}$ correspondence is limited to the Torus and Möbius band. This is a conditional incidental observation with no claim of theoretical necessity.

Appendix D. Sensitivity Analyses and Experiment Overview

AU threshold δ : AU values are essentially unchanged for $\delta \in [10^{-3}, 10^{-1}]$ (MNIST FC-VAE, 300 epochs).

β **sensitivity**: in preliminary experiments (50 epochs, $\beta \in \{1, 2, 4, 8\}$), $\beta = 1$ gives AU $\approx m$ (under-regularization) and $\beta = 8$ shows signs of over-regularization; AU patterns are stable for $\beta \in [2, 8]$. The default $\beta = 4$ of this paper is the center of this stable range.

k **sensitivity of MLE**: MLE = 11.5–12.4 (MNIST) for $k \in \{5, 10, 20\}$, stable.

Table A4. Experiment overview. Main-text VAE verification results used for the principal verdicts use 3 seeds (42, 123, 777); single-seed sweeps are marked in the Seeds column; appendix experiments marked “single” use seed 42 only.

Experiment	Data / model	Epochs	Seeds	Section
Exp-Syn-1 / Exp-Syn-AU	Swiss Roll, Torus / AE, VAE	200–300	3	Section 5.2
Exp-Syn-Verify	Torus, Swiss Roll / VAE ($m = 3\text{--}8$)	200	5	Section 5.2
Exp-Mnist-Base	MNIST / AE (m_{ref} sweep)	300	1 (sweep)	Section 5.3
Exp-Mnist-Multi-300	MNIST / FC-VAE	300	3	Section 5.3
Exp-Mnist-Dense-300	MNIST / FC-VAE (dense grid)	300	3	Section 5.3.5
Exp-Boot-CI	MNIST / ref.-AE latents (re-analysis)	—	subsamp.	Section 5.3.2
Exp-Downstream	MNIST / AE probe, FC-VAE probe	300	1 (AE), 3 (VAE)	Section 5.4
Exp-Real-Geom	MNIST, F-MNIST / AE, IsoAE, CAE, DAE	300	3	Section 5.5
Exp-Conv-M256 / M512	MNIST / Conv-VAE	150	3	Section 5.6
Exp-Nat-Cifar / Exp-Nat-SVHN	CIFAR-10, SVHN / Conv-VAE	150	3	Section 5.6
Exp-Fashion-MS	F-MNIST / FC-VAE	300	3	Section 5.3.6
Exp-ConvV-HiBeta	MNIST / Conv-VAE (β_{max} up)	150	3	App. C
Exp-NatImg-Sweep	CIFAR-10 / Conv-VAE	300	partial	App. C
Exp-Fashion	F-MNIST / AE, VAE (preliminary)	100	single	App. C
Exp-Syn-Top	Torus, Möbius, S^1 / VAE	300	single	App. C

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